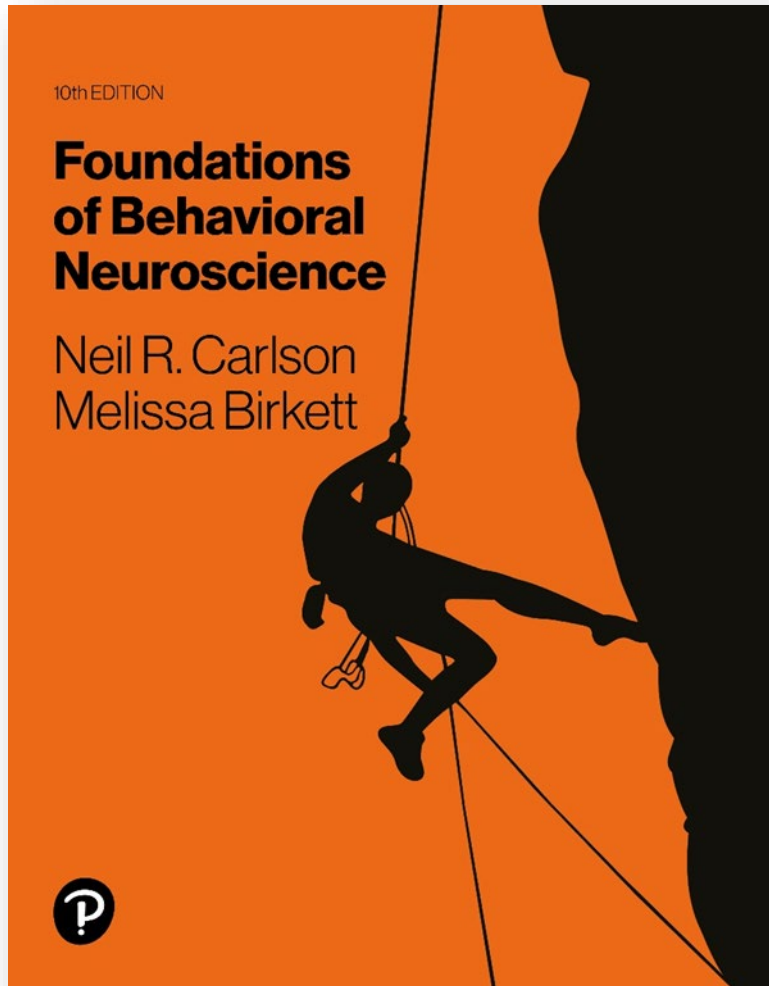


Foundations of Behavioral Neuroscience

Tenth Edition



Chapter 14

Human Communication

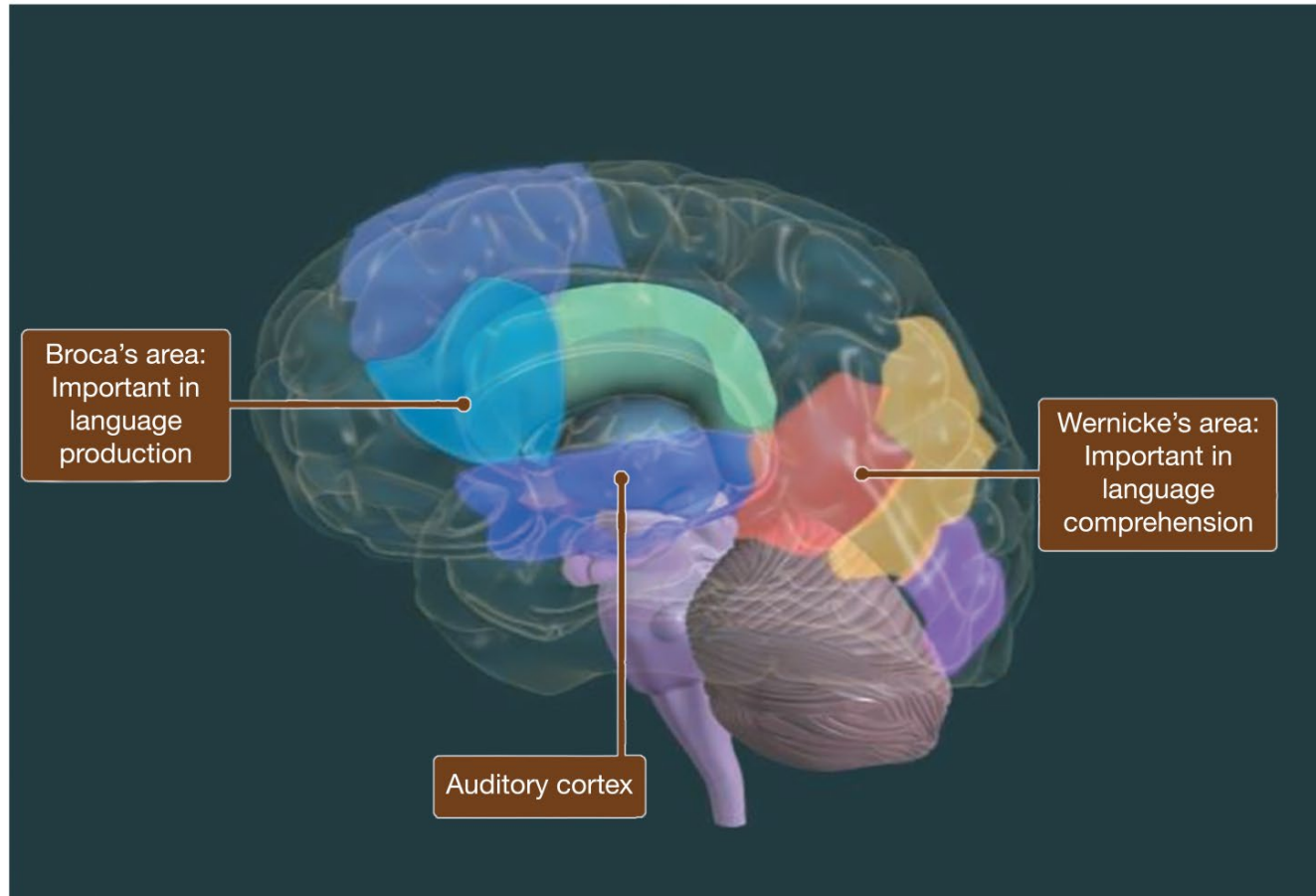
Language Production and Comprehension: Brain Mechanisms

- Lateralization
- Language Production
- Language Comprehension
- Bilingualism
- Prosody
- Recognition of People's Voices

Language Disorders

- **Aphasia** is a disturbance in the comprehension or production of language
 - Broca's aphasia
 - Wernicke's aphasia
- Not due to sensory or motor dysfunction

Figure 14.1 Language Areas

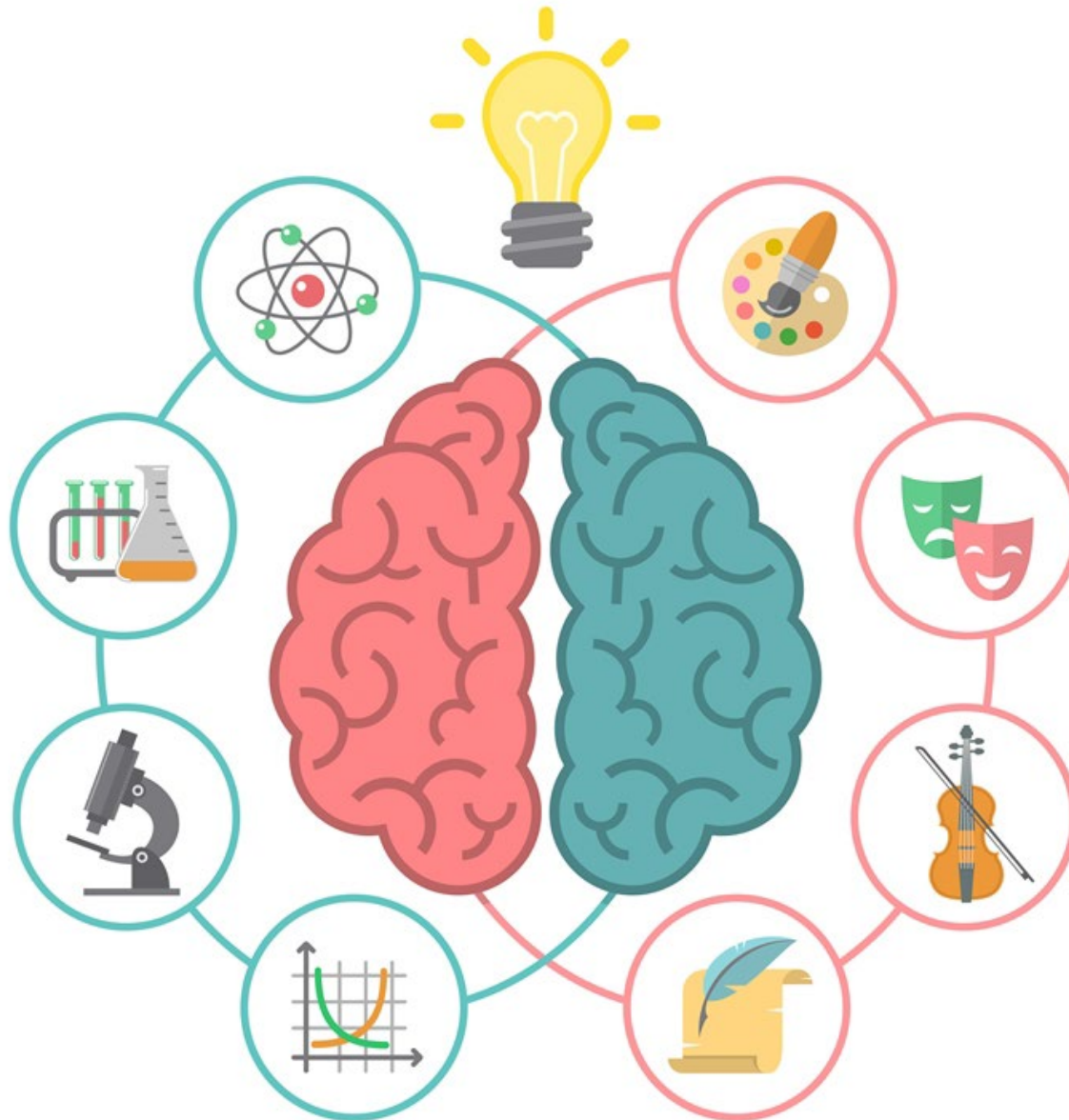


This figure shows two important language areas of the brain.

Lateralization (2 of 2)

Both hemispheres are involved in language

- **Left hemisphere** typically controls comprehension, analysis of speech, and production
- **Right hemisphere** controls activities like reading maps, perceiving spatial relations, recognizing complex geometrical forms, organizing a narrative, the expression and recognition of emotion in the tone of voice, and prosody



Lateralization (2 of 2)

90 percent of the population uses the left hemisphere for language

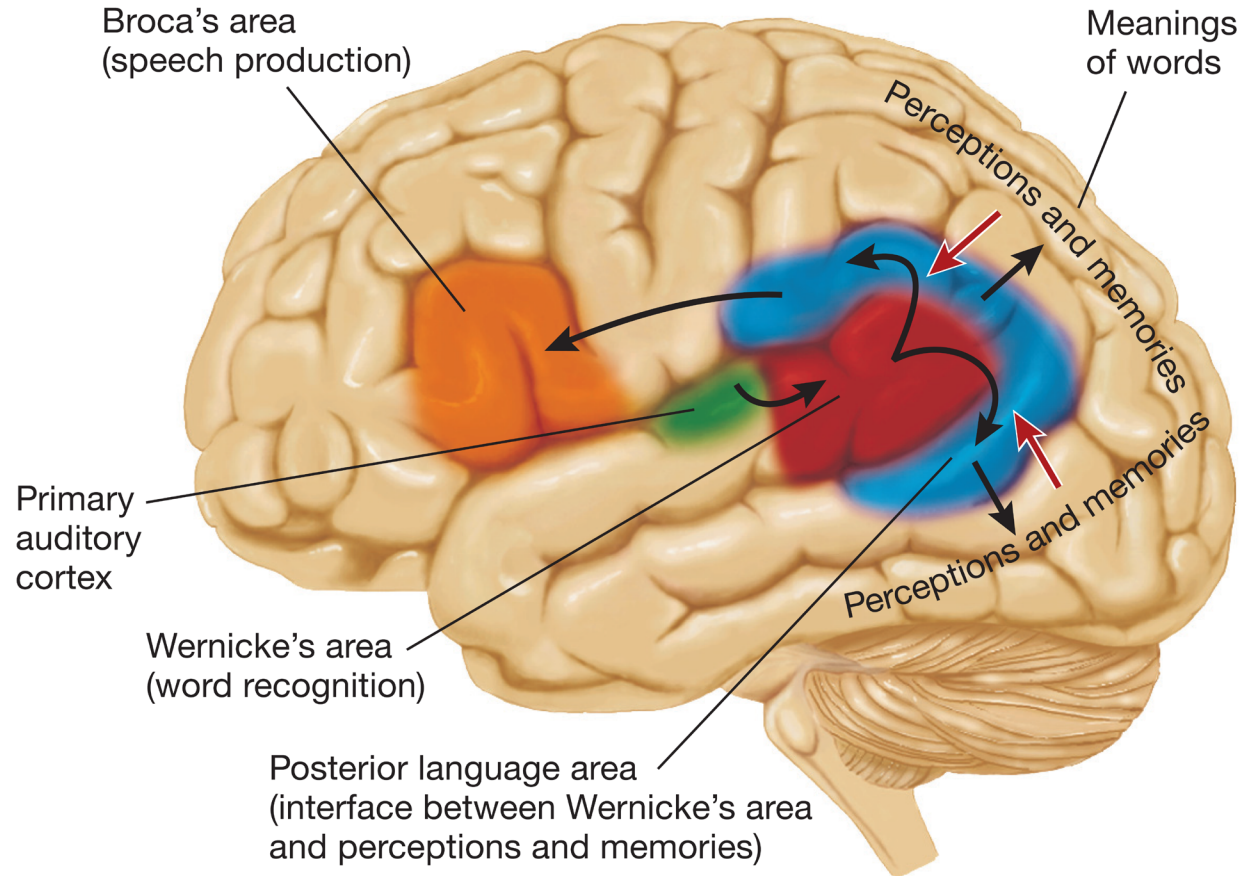
Right hemisphere speech is only found in

4% of those who are right-handed

15% of those who are ambidextrous

27% of those who are left-handed

Figure 14.2 The Dictionary in the Brain



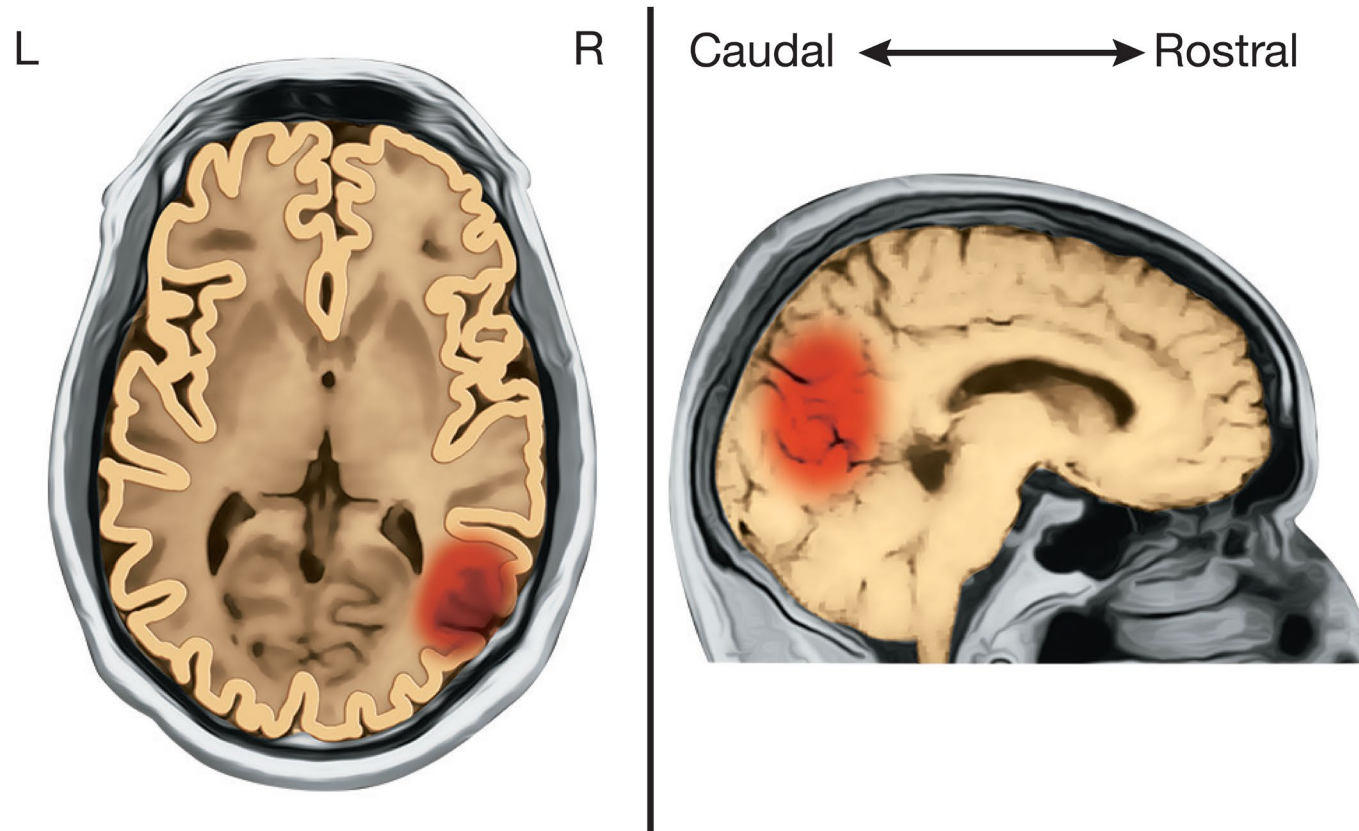
Wernicke's area contains the auditory entry of words; the meanings are contained as memories in the sensory association areas. Black arrows represent comprehension of words—that is, the activation of memories that correspond to a word's meaning. Red arrows represent translation of thoughts or perceptions into words.

Language Comprehension

Factors in Comprehension

- Comprehension of figurative aspects of language involves right hemisphere
 - Metaphors, proverbs, moral stories

Figure 14.3 Evaluating Metaphors



These images of neural activity were produced by evaluating the meaning of metaphors. Source: (Based on Sotillo, M. et al., *Neuroscience Letters*, 2005, 373, 5–9.) (Based on Sotillo, M., Carretié, L., Hinojosa, J. A., Tapia, M., Mercado, F., López-Martín, S., & Albert, J. (2004). Neural activity associated with metaphor comprehension: spatial analysis. *Neuroscience letters*, 373(1), 5–9.)

Metaphors

Learning Objective 14.4 Describe evidence for common and language-specific brain regions for bilingual language processing.

Nichelli et al., 1995 found that judging the moral of Aesop's fables activated regions of the right hemisphere

Pobric et al., 2008 found that using TMS on the right superior temporal cortex impaired the person's ability to understand novel metaphors such as "conscience storm" but kept conventional metaphors such as "sweet voice" intact.

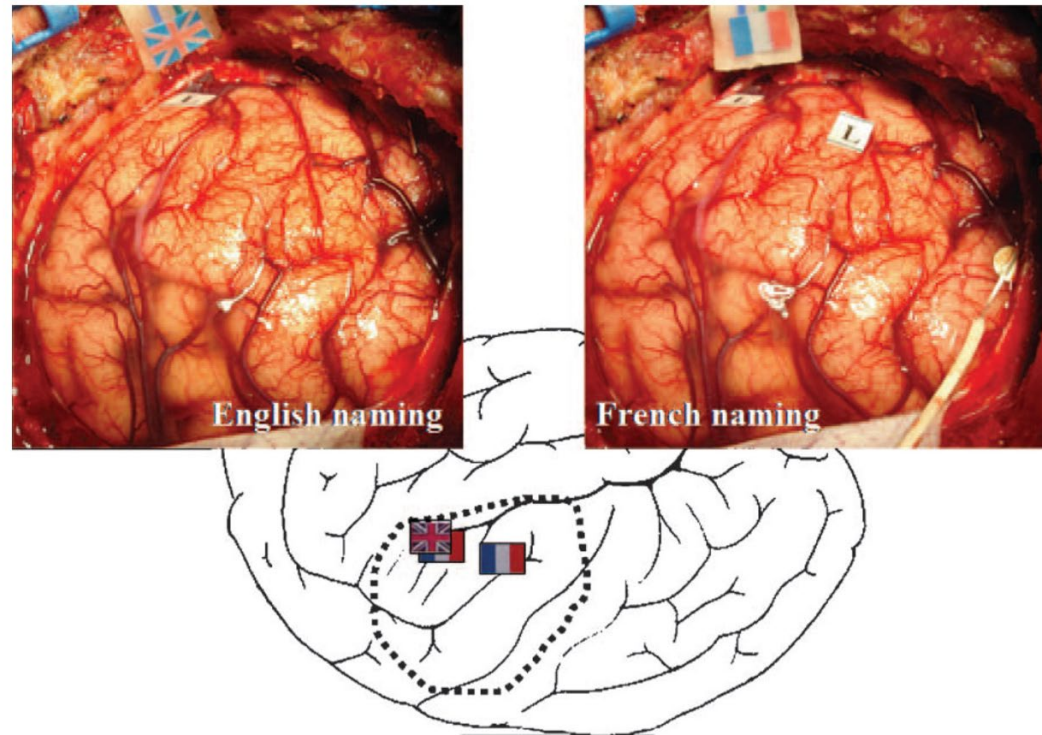
Bilingualism

Majority of people are bilingual or multilingual

Some brain regions in bilingual individuals are devoted to specific languages and some common to both languages; identifiable using imaging studies

- Common and language-specific areas found in the frontal and posterior temporal/parietal cortexes
- Language-specific areas found in some subcortical structures

Figure 14.4 Mapping Common and Language-Specific Areas of the Cortex



A 31-year-old bilingual English-French right-handed patient was operated on for a small tumor located in the left parietal lobe. Direct cortical electrostimulation revealed one French-specific naming site (single flag) whereas one site was common for English and French (overlapping flags), confirming different common and language-specific regions.

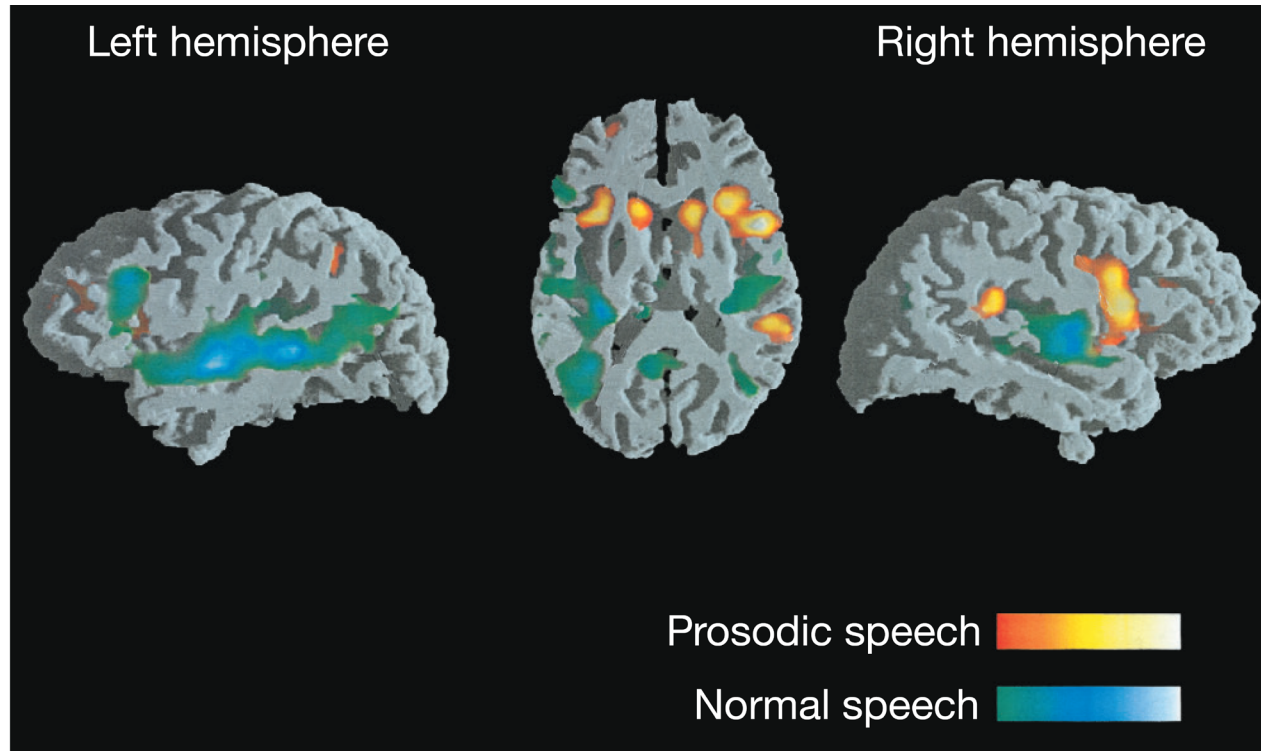
Source: (Giussani, C., Roux, F. E., Lubrano, V., Gaini, S. M., et al. Review of language organisation in bilingual patients: What can we learn from direct brain mapping? *Acta Neurochirurgica*, 2007. 149(11), 1109–111.)

Prosody

Brain Structures and Functions

- **Prosody** refers to rhythmic, emphatic, and melodic aspects of speech
 - Function of the right hemisphere
 - Likely related to musical skills, expression and recognition of emotions
- Broca's aphasia (left hemisphere damage) can also cause deficits in prosody

Figure 14.5 Listening to Normal Speech or Its Prosodic Components



Functional MRI scans were made while participants listened to normal speech (blue and green regions) or the prosodic elements of speech with the meaningful components filtered out (orange and yellow regions). Source: (From Meyer, M., Alter, K., Friederici, A. D., Lohmann, G., and von Cramon, D. Y., FMRI reveals brain regions mediating slow prosodic modulations in spoken sentences, *Human Brain Mapping*, 2002, 17, 73–88. Reprinted with permission.)

Recognition of People's Voices

Brain Structures and Functions

- We learn at early age to recognize voices
- Regions of the right hemisphere are involved in recognizing specific voices
 - Phonagnosia caused by damage to these areas

Disorders of Language Production and Comprehension

- Disorders of Language Production: Broca's Aphasia
- Disorders of Language Comprehension: Wernicke's Aphasia
- Conduction Aphasia
- Aphasia in People Who Are Deaf
- Stuttering

Disorders of Language Production: Broca's Aphasia

Broca's Aphasia

- Caused by damage to frontal lobe: produces **agrammatism, anomia, articulation difficulties**
 - Can comprehend language better than produce it
 - Difficulty in using **function words** (a, the, some)
 - Reliance on **content words** (apple, house)
 - Relies on circumlocution
- Can also be caused by lesions of the basal ganglia

Content words vs function words

Content words carry the main meaning in a sentence, including nouns, main verbs, adjectives, and adverbs.



Function words, also known as grammatical words, provide structure and grammatical relationships, including prepositions, pronouns, conjunctions, and auxiliary verbs



Circumlocution: a roundabout way of speaking

what are other
words for
circumlocution?

periphrasis, wordiness,
verbiage, redundancy, prolixity,
tautology, verbosity, pleonasm,
euphemism, diffuseness



 Thesaurus.plus

Figure 14.6 Assessment of Aphasia



The drawing of the kitchen story is part of the Boston Diagnostic Aphasia Test.

Disorders of Language Production: Broca's Aphasia

- Includes damage to underlying subcortical white matter, bundles of axons
- Motor memories-of the sequences of muscular movements that are needed to articulate words
- Basal ganglia involvement
- Broca's area has projections to primary motor cortex

Speech Production

Agrammatism One of usual symptoms of Broca's aphasia

A difficulty in comprehending or properly employing grammatical devices, such as verb endings and word order
Use –ed, have, add -ing

Anomia Difficulty in finding (remembering) appropriate word to describe object, action, or attribute; one of symptoms of aphasia.

Speech production: Articulation Difficulties

- Individuals with Broca's aphasia also have difficulty in articulation
 - Pronounce lipstick by saying likstip

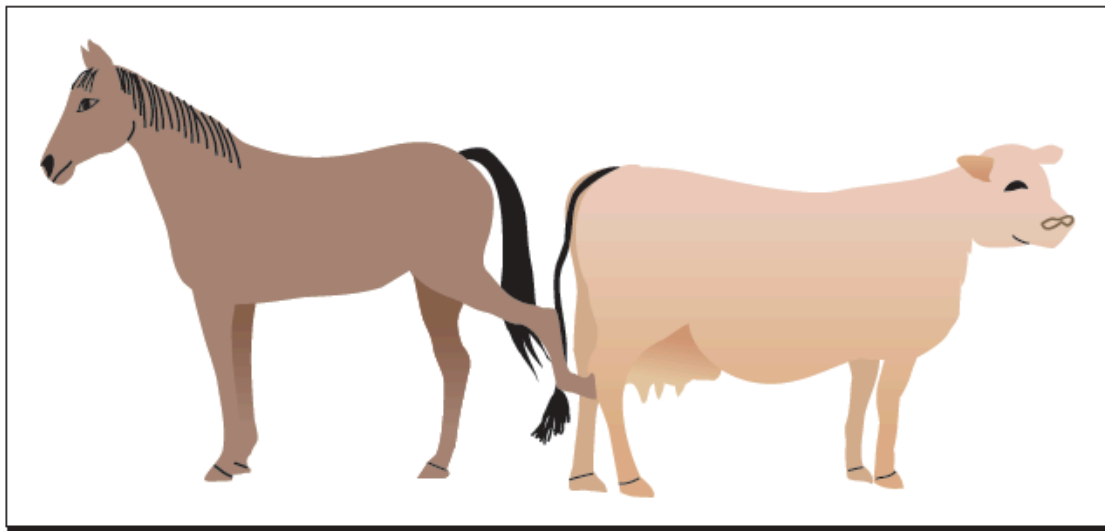
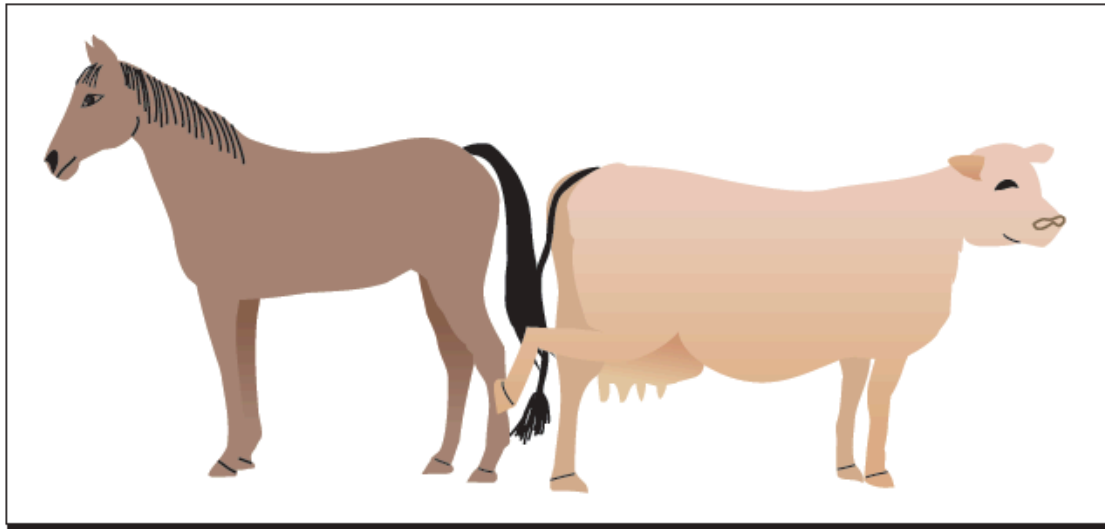
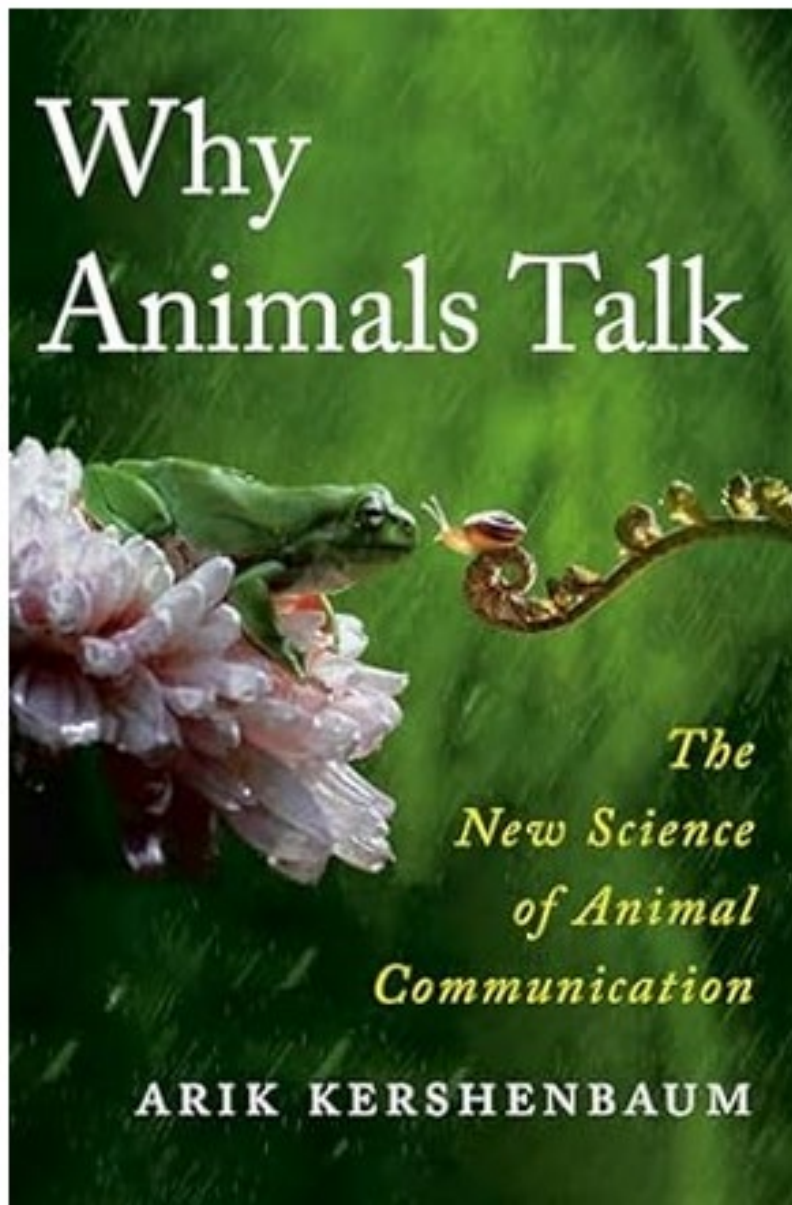


FIGURE 13.3 Assessment of Grammatical Ability. This is an example of the stimuli used in the experiment by Schwartz, Saffran, and Marin (1980).

Cow and horse

- The subject's task was to point to the appropriate picture
 - The horse kicks the cow
- They performed very poorly
- Why is grammar so important?



Disorders of Language Comprehension: Wernicke's Aphasia

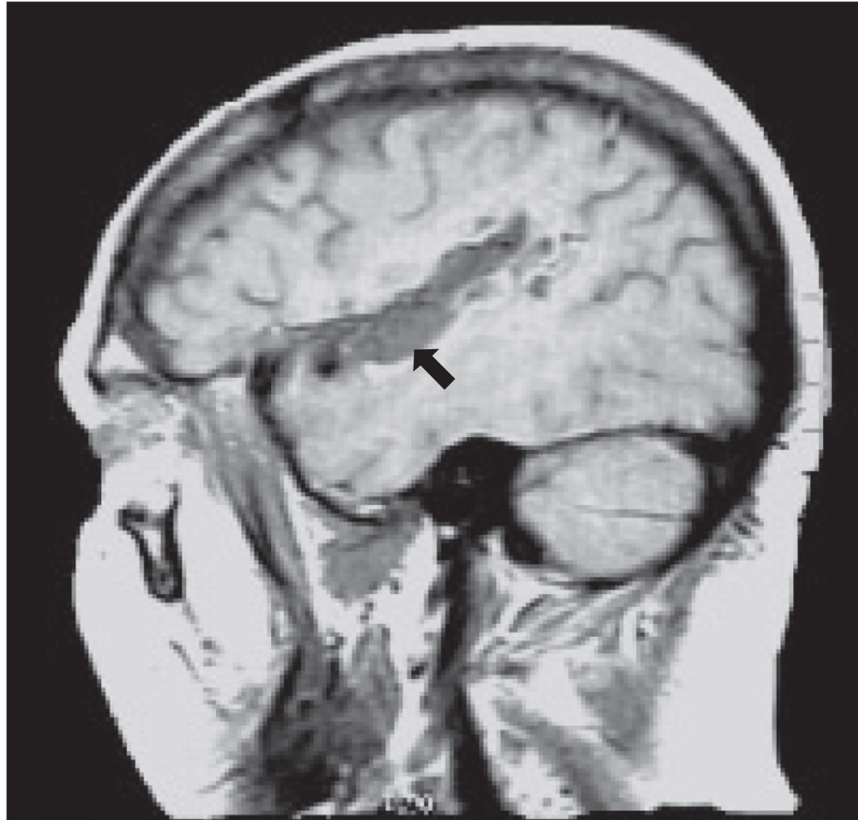
What are some of the characteristics of Wernicke's Aphasia?

Disorders of Language Comprehension: Wernicke's Aphasia

Wernicke's Aphasia

- Receptive aphasia caused by damage to Wernicke's area in the left hemisphere and the posterior language area
 - Afflicted individuals seem unaware of deficits
- Poor language comprehension and production of meaningless speech.

Pure Word Deafness: Recognition and comprehension are not the same



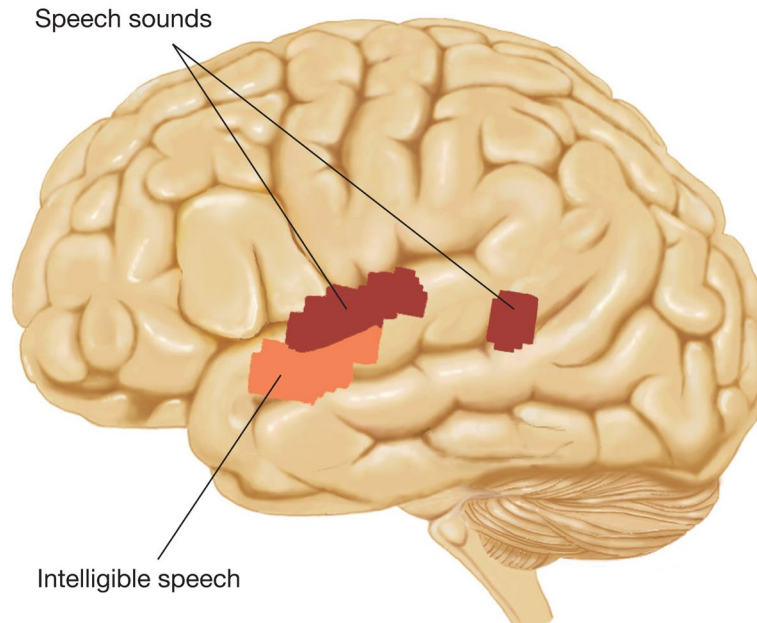
The listener knows you are telling them words but they do not understand them

Pure Word Deafness:

- They can perceive and recognize speech but do not understand the words
- Can recognize emotion
- Can still understand the speaker if they read their lips
- Their own speech is unaffected
- Can still read and write words

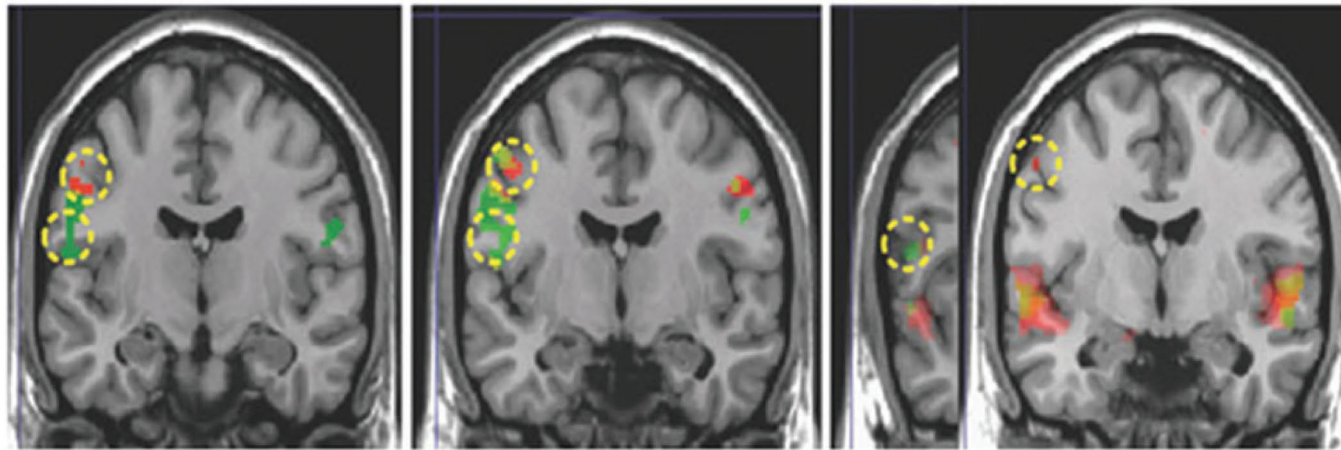
Figure 14.8

Responses to Speech Sounds



Results of PET scans indicate the regions of the superior temporal lobe that respond to speech sounds. **Red:** Regions that responded to phonetic information (normal speech sounds or a computerized transformation speech that preserved the complexity of the speech sounds but rendered it unintelligible). **Orange:** Region that responded to intelligible speech (normal speech sounds or a computerized transformation that removed most normal frequencies but preserved intelligibility).

Figure 14.9 Mirror Neurons and Speech



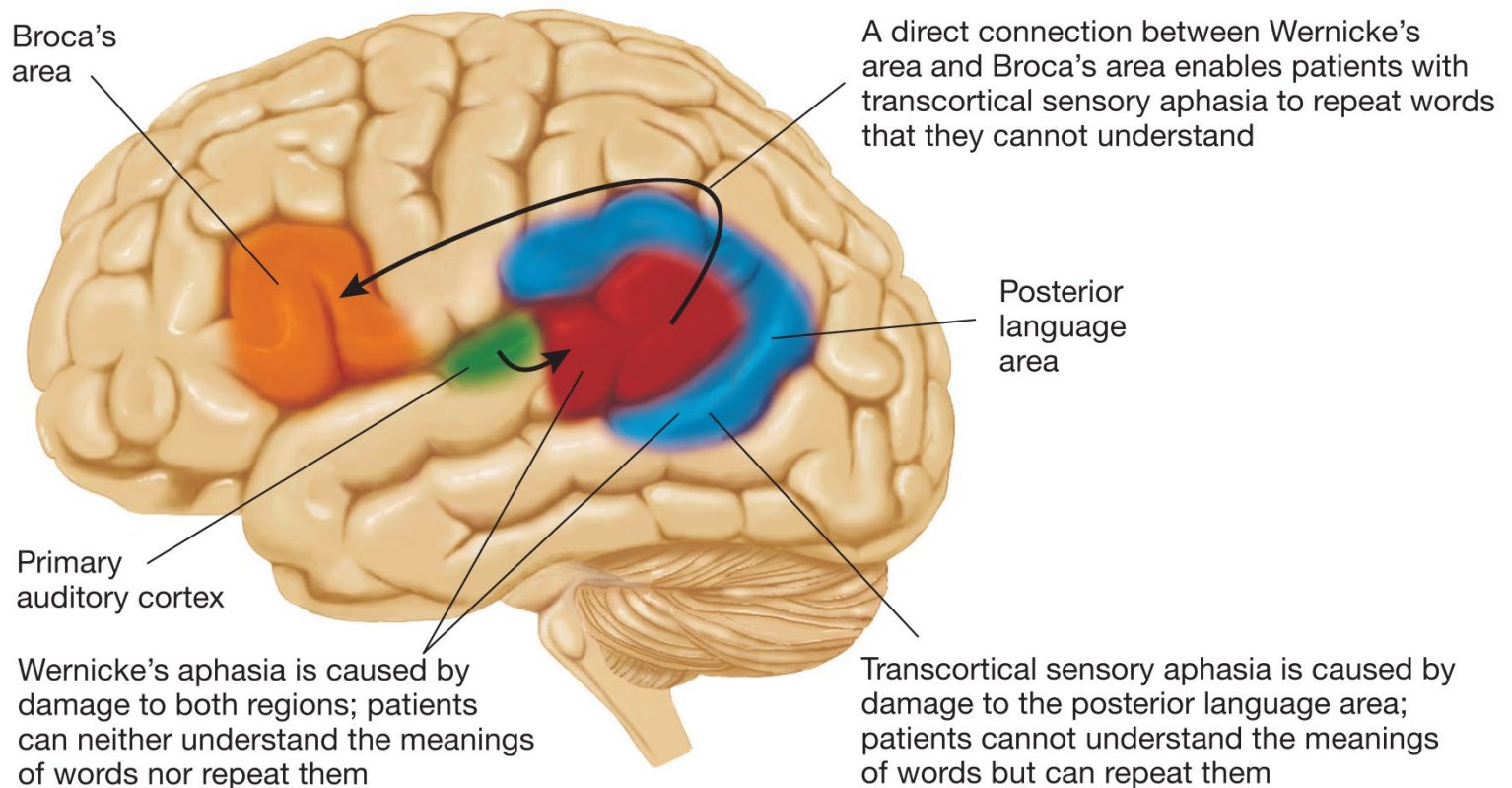
Said syllables aloud

Said syllables
to themselves
silently

Heard syllables said

Brain activation in the primary motor cortex that occurred when people said syllables aloud, said syllables to themselves silently, or heard someone else saying syllables. The two regions marked with yellow circles are involved in control of tongue movements (syllables that included the sound of t, shown in green) and lip movements (syllables that included the sound of p, shown in red). Source: (From Pulvermüller, F., Huss, M., Kherif, F., et al., Motor cortex maps articulatory features of speech sounds, *Proceedings of the National Academy of Sciences, USA*, 2006, 103, 7865–7870. Reprinted with permission.)

Figure 14.10 Transcortical Sensory Aphasia and Wernicke's Aphasia



This schematic shows the location and interconnections of the posterior language area and an explanation of its role in transcortical sensory aphasia and Wernicke's aphasia.

Transcortical Sensory Aphasia

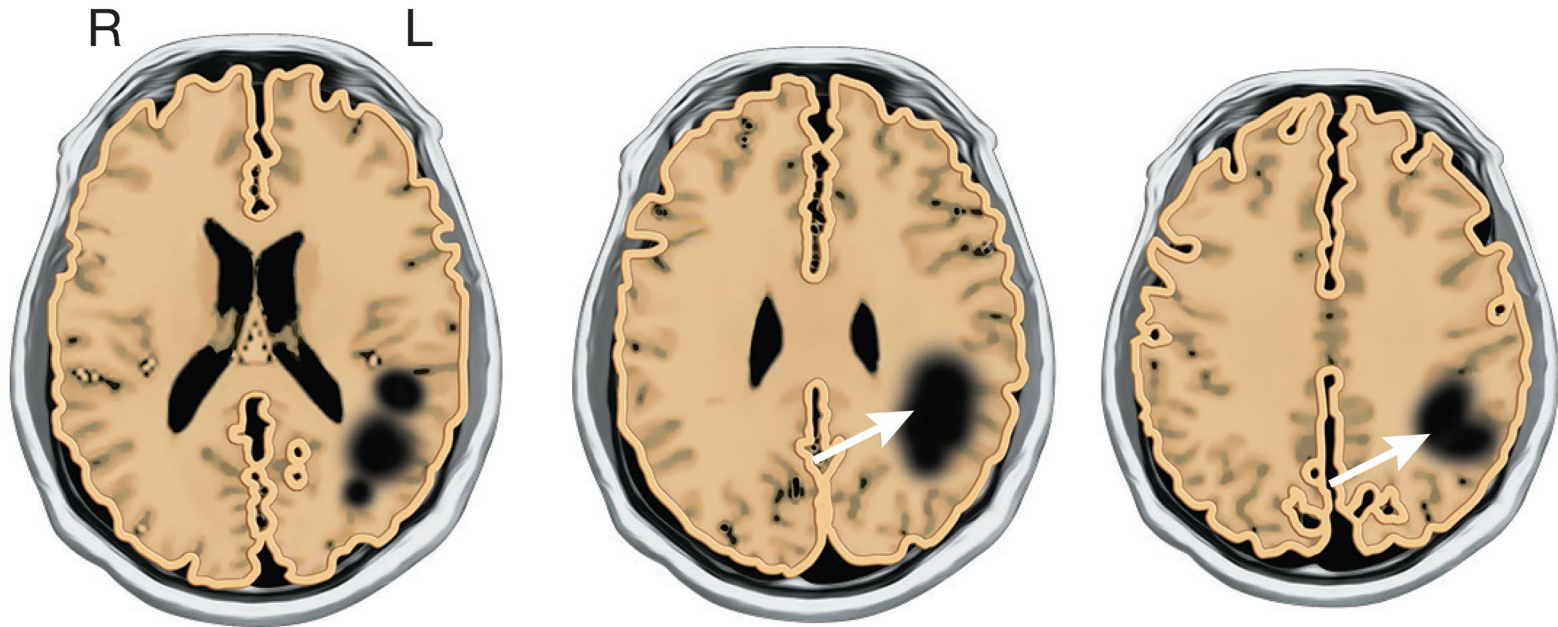
- They can repeat what is being said to them but they do not understand what they are hearing
- Cannot produce meaningful speech of their own



Conduction Aphasia

- Fluent and meaningful speech with good comprehension, but very poor repetition
- Cannot repeat **Nonwords**
- Caused by damage to arcuate fasciculus
 - Suggests direct connection between Wernicke's area and Broca's area
- A second pathway, between the posterior language area and Broca's area, is indirect and is based on the *meaning* of words, not on the sounds they make.

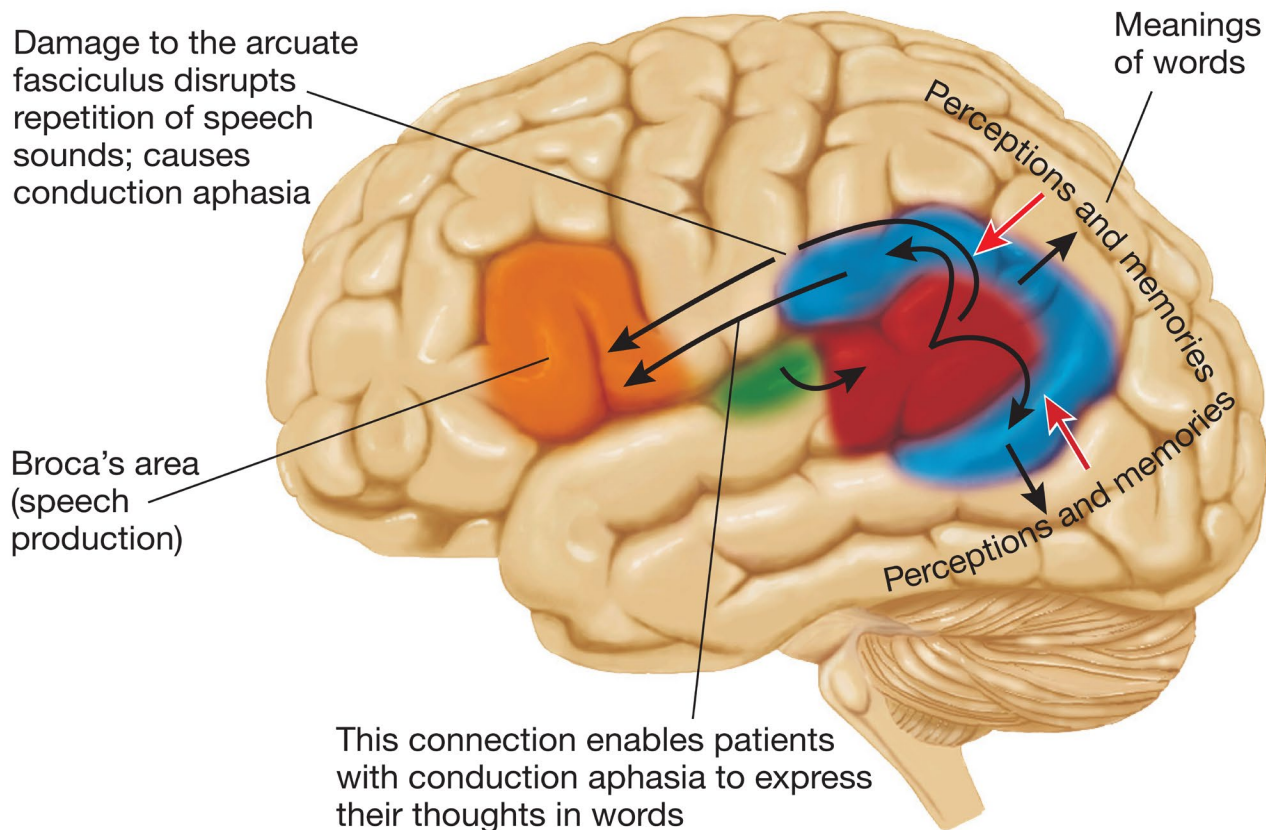
Figure 14.12 Conduction Aphasia



MRI scans show the subcortical damage responsible for a case of conduction aphasia. The lesion damaged the arcuate fasciculus, a fiber bundle connecting Wernicke's area and Broca's areas. Source: (Based on Arnett, P. A., Rao, S. M., Hussain, M., Swanson, S. J., and Hammeke, T. A. *Neurology*, 1996, 47, 576–578.)

Figure 14.13

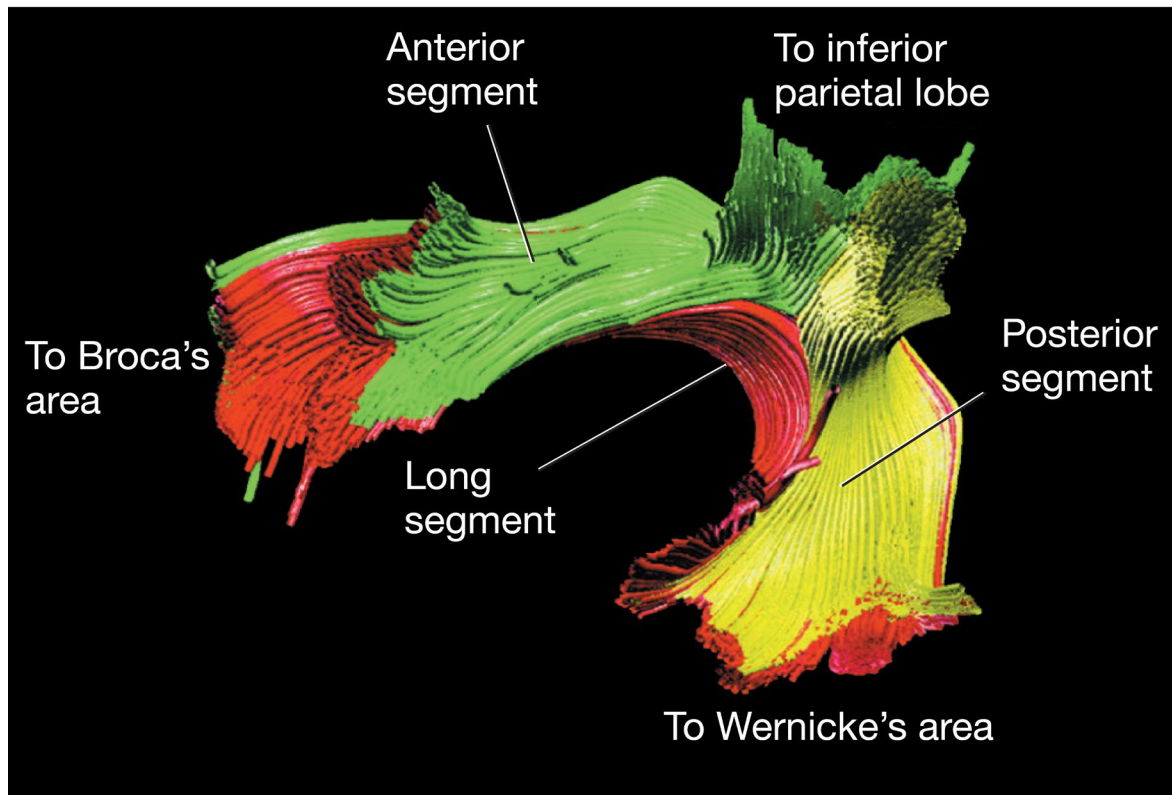
Explanation of Conduction Aphasia



A lesion that damages the arcuate fasciculus disrupts the transmission of auditory information, but not information related to meaning, to the frontal lobe.

Figure 14.14

Components of the Arcuate Fasciculus



A computer-generated reconstruction of the components of the arcuate fasciculus was obtained through diffusion tensor MRI. Source: (From Catani, M., Jones, D. K., and ffytche, D. H., Perisylvian language networks of the human brain, *Annals of Neurology*, 2005, 57, 8–16. Reprinted with permission.)

Broca's area disruption

- Aziz-Zadeh et al., 2005 used TMS to Broca's area while people were silently counting the number of syllables in words presented on a screen.
- Using TMS they disrupted overt (actual) speech

Arcuate Fasciculus disruption

- They also found that TMS disrupted covert (non-verbalized) speech.
- The participants took longer to count with TMS.

Arcuate fasciculus

Learning Objective 14.9 Identify the symptoms of conduction aphasia, and describe their contribution to understanding about connections between Broca's and Wernicke's areas.

The arcuate fasciculus is responsible for a phonological loop.

Important role in short-term memory of words and speech sounds one just heard.

Compare transcortical sensory aphasia and conduction aphasia

Someone who has transcortical sensory aphasia would be able to repeat what someone else said.

Someone with conduction aphasia is unable to repeat nonwords.

Aphasia in People Who Are Deaf

Sign Language

- Expressed by movements of the hands; most common form is American Sign language (ASL)
- Left hemisphere plays more important role (as it does in spoken language)
- Some believe sign language precursor to spoken language
- Role of mirror neurons

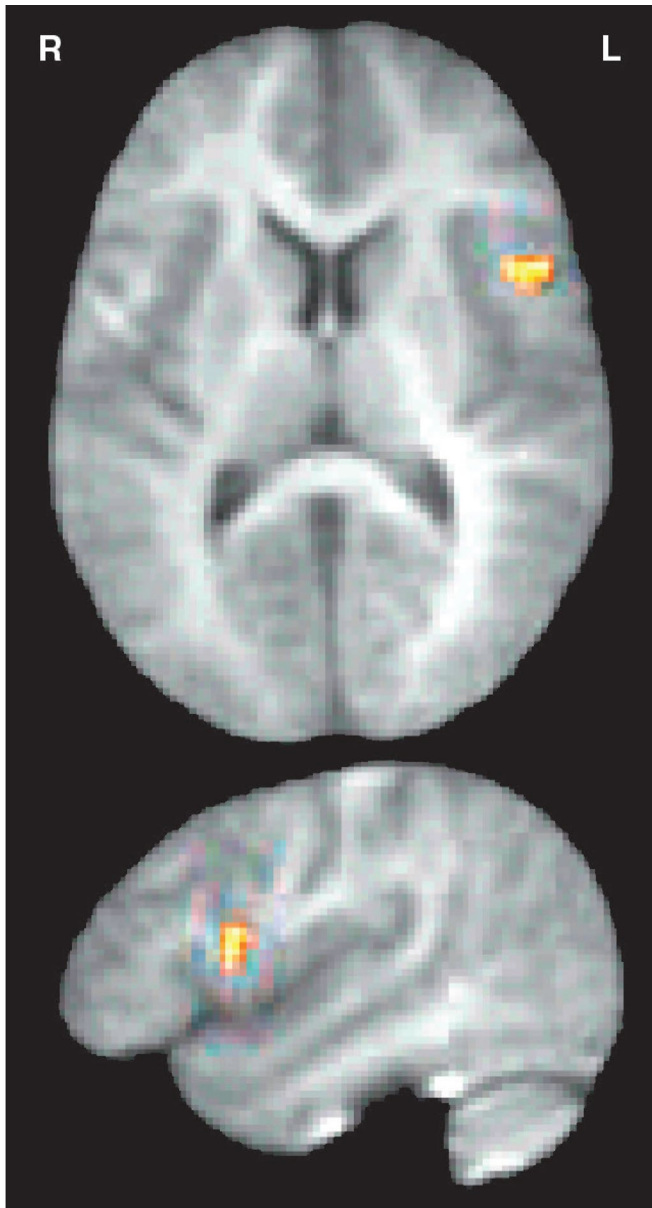
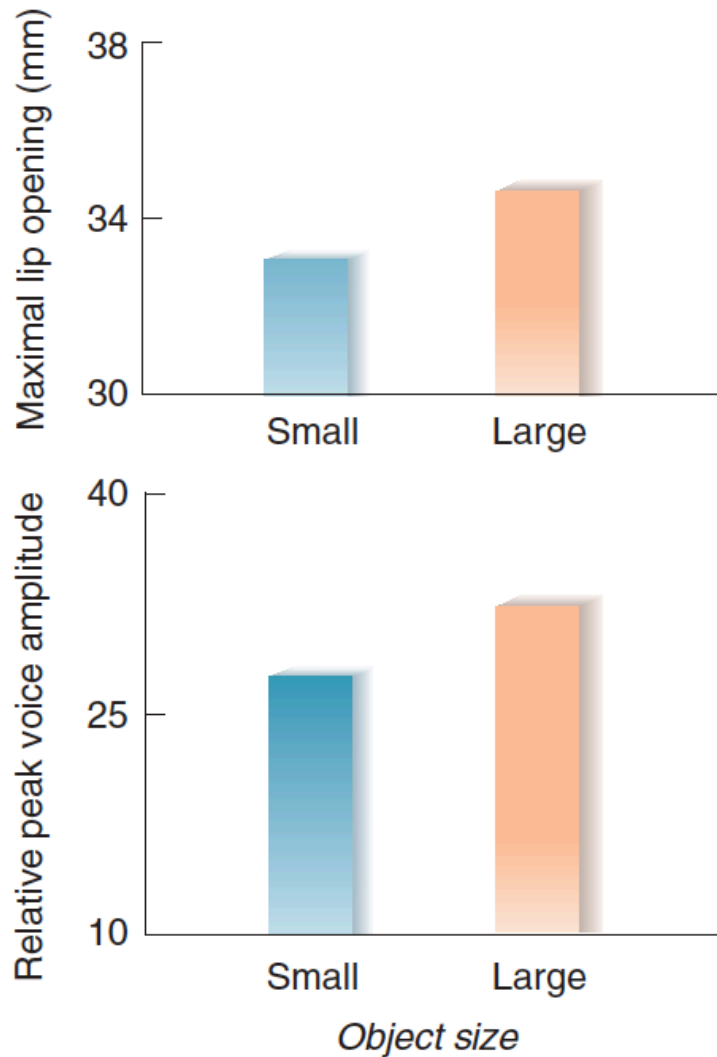


Figure 14.15

Mirror Neurons in Broca's Area

PET scans show a region of the inferior left frontal lobe that was activated when a person saw a finger movement or imitated it. (top) Horizontal section. (bottom) Lateral view of left hemisphere. Source: (From Iacoboni, M., Woods, R. P., Brass, M., Bekkering, H., Mazziotta, J. C., and Rizzolatti, G., Cortical mechanisms of human imitation, *Science*, 1999, 286, 2526–2528. Copyright © 1999 by the American Association for the Advancement of Science. Reprinted with permission.)



Repeat ba or ga
while watching someone grasp for
different sized objects

The region of the brain that controls
grasping also controls speech
movements

FIGURE 13.15 Links Between Hand and Mouth. The graphs show lip opening and voice amplitude of subjects repeating syllables while watching someone grasping small and large objects.

Aphasia in People Who Are Deaf

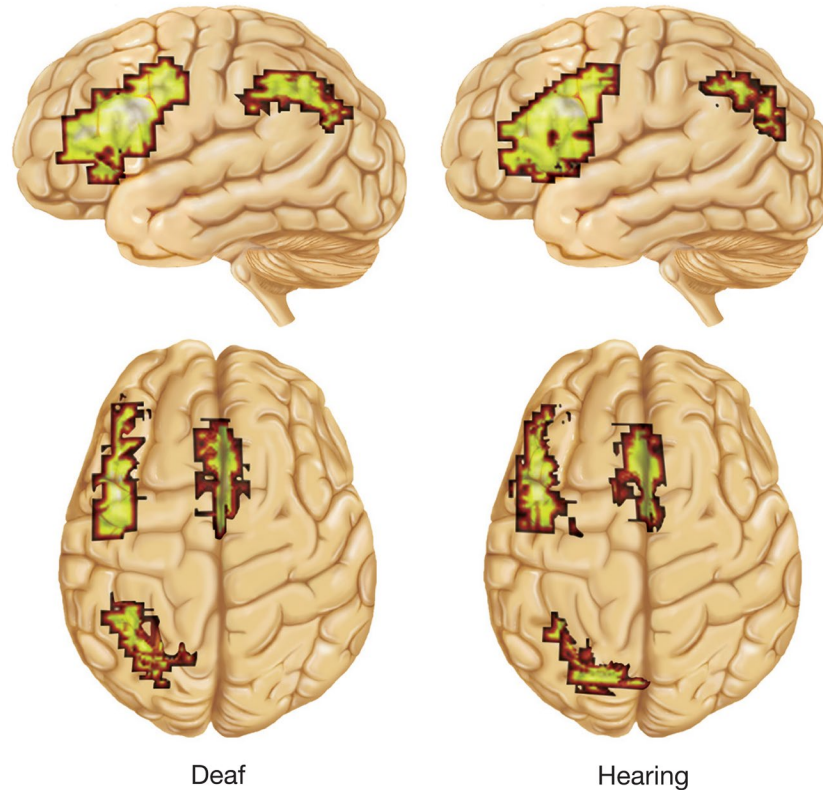
Learning Objective 14.10 Explain the brain processes involved in production and comprehension of American Sign Language.

Several studies have found a linkage between speech and hand movements

Grammar for ASL is based on spatial nature

Maybe the right hemisphere has stronger role in grammar but unknown.

Figure 14.16 Brain Activation During a Rhyming Task



The same language-related regions of the brain are activated in people who are deaf or have typical hearing when they decide whether two written words rhyme. Source:

(MacSweeney, M., Waters, D., Brammer, M. J., Woll, B., & Goswami, U. (2008). Phonological processing in deaf signers and the impact of age of first language acquisition. *Neuroimage*, 40(3), 1369–1379.)

Stuttering

Learning Objective 4.11 Describe the biological basis for stuttering and identify treatment strategies.

What are some of the characteristics of stuttering?

Stuttering

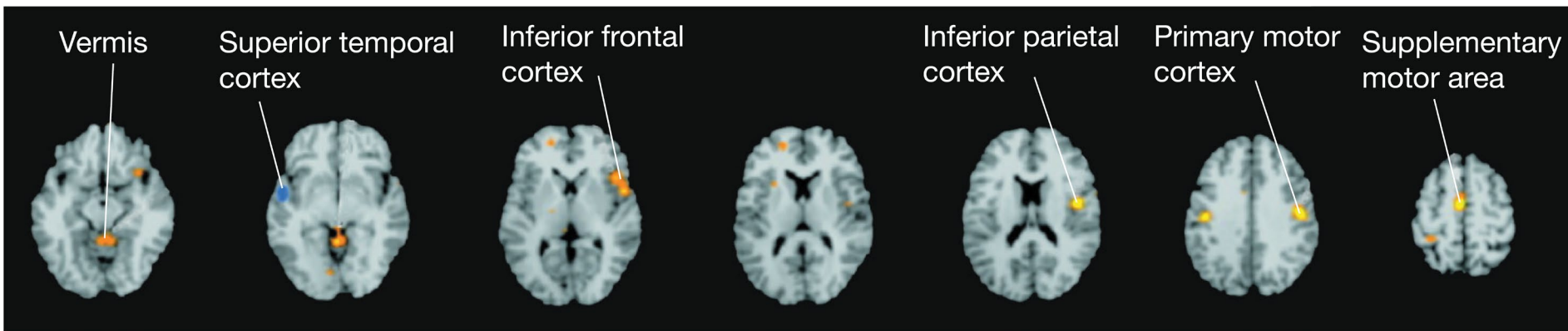
- Characterized by pauses, prolongation of sounds, or repetitions of sounds, syllables or words that disrupt normal flow of speech
 - Influenced by genetic factors
 - Abnormal auditory feedback
- Caused by abnormalities in neural mechanisms involved in planning and initiating speech

Speech Production and Comprehension: Brain Mechanisms

Stuttering

Appears to be influenced by genetic factors and affects approximately 1 percent of population; three times more prevalent in men than in women (Brown et al. 2005; Fisher 2010)

Figure 14.17 Stuttering and the Brain



Computer-generated images show data from a meta-analysis of functional-imaging studies of fluent speakers (blue regions) and people who stutter (orange regions), taken while they were speaking. Source: (From Brown, S., Ingham, R. J., Ingham, J. C., Laird, A. R., and Fox, P. T., Stuttered and fluent speech production: An ALE meta-analysis of functional neuroimaging studies, *Human Brain Mapping*, 2005, 25, 105–117. Reprinted with permission.)

Stuttering

Neumann et al. (2005)

- Provide further evidence that apparently abnormal auditory feedback in stutterers is reflected in decreased activation of temporal cortex



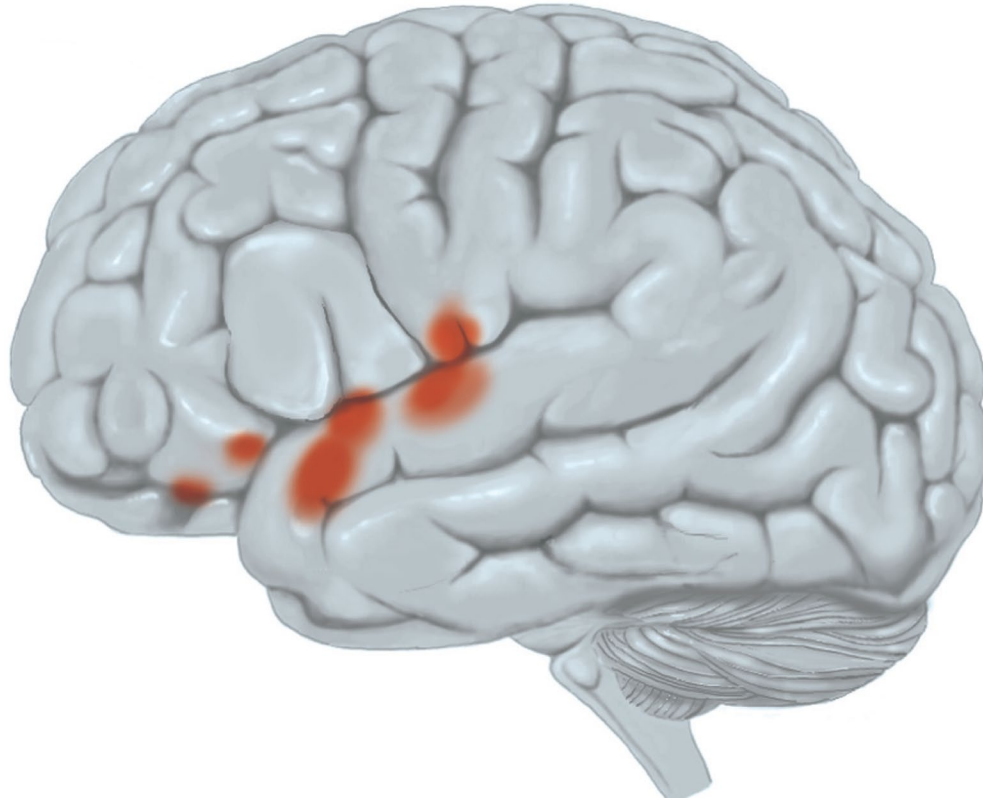
Stuttering

Neumann et al. (2005)

- fMRI used to measure regional brain activation of stutterers reading sentences aloud during two sessions, one before and one after a successful twelve-week course of fluency shaping therapy
- Figure 13.18 shows that after the therapy, activation of temporal lobe was increased

Figure 14.18

Effects of Therapy for Stuttering



A functional MRI scan shows regions of the superior temporal lobe that showed increased activity one year after a successful course of therapy for stuttering. Source: (Based on Neumann, K., Preibisch, C., Euler, H. A., Von Gudenberg, A. W., Lanfermann, H., Gall, V., & Giraud, A. L. (2005). Cortical plasticity associated with stuttering therapy. *Journal of fluency disorders*, 30(1), 23–39.)

Disorders of Reading and Writing

- **Relation to Aphasia**
- **Pure Alexia**
- **Toward an Understanding of Reading**
- **Toward an Understanding of Writing**

Relation to Aphasia

Learning Objective 14.12 Compare aphasias and disorders of reading and writing.

Reading and Writing

- Brain damage can also produce reading and writing disorders
- Aphasias are usually accompanied by parallel reading and writing deficits

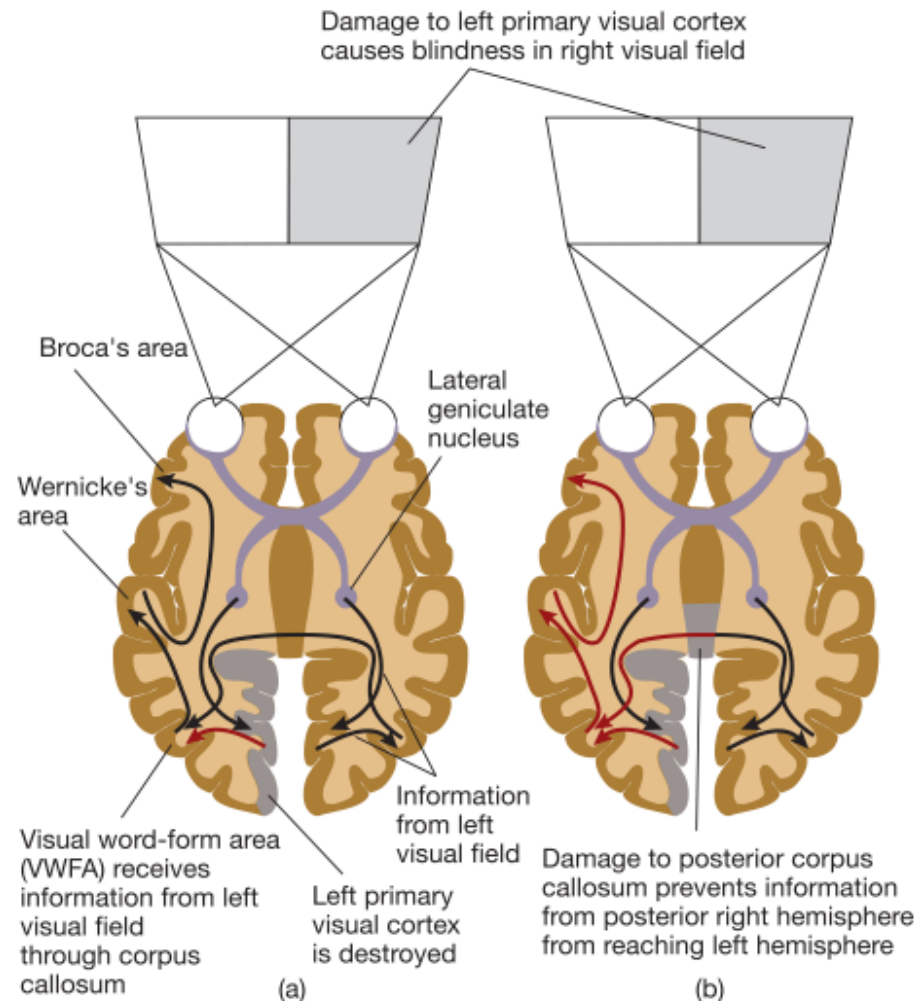
Pure Alexia

Learning Objective 14.13 Identify the symptoms of pure alexia, and describe the brain structures involved.

What are the Symptoms and brain areas impacted for pure alexia?

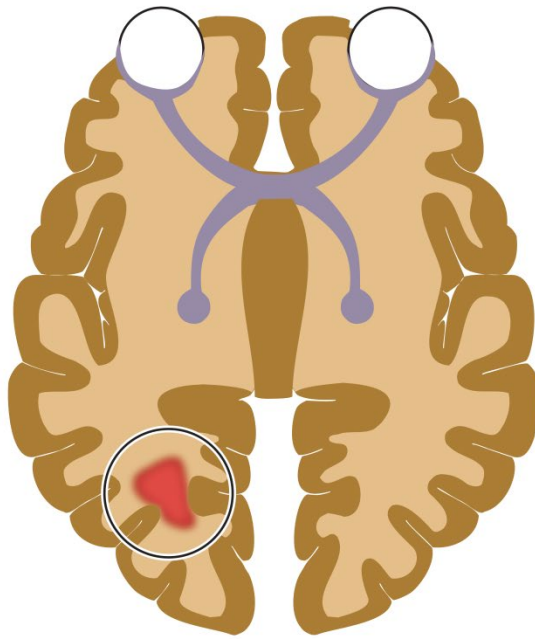
Pure Alexia: Symptoms

- Perceptual disorder characterized by loss of ability to read without loss of the ability to write; produced by brain damage
 - Can recognize words spelled aloud
- Visual information prevented from reaching visual association cortex
 - Patients have difficulty with visual, not auditory, input
- Damage to left visual cortex and posterior corpus callosum
- Similar to Pure Word Deafness

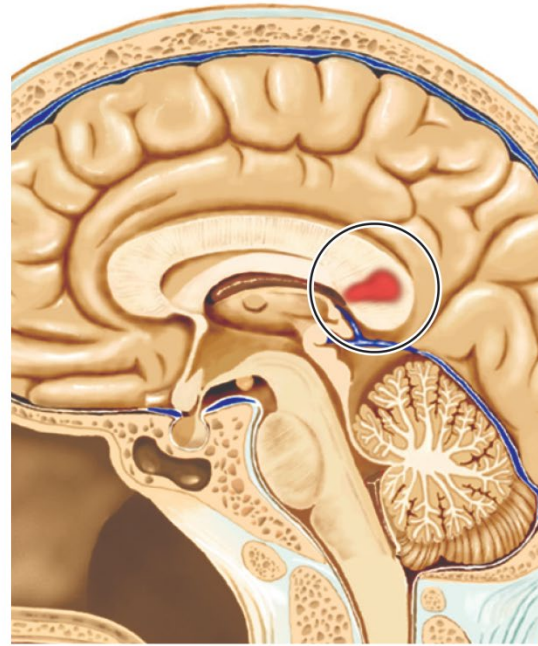


In this schematic, red arrows indicate the flow of information that has been interrupted by brain damage. (a) The route followed by information as a person with damage to the left primary visual cortex reads aloud. (b) Additional damage to the posterior corpus callosum interrupts the flow of information and produces pure alexia.

Figure 14.20 Pure Alexia in a Person with Multiple Sclerosis



Damage to white matter that serves the left visual cortex



Damage to posterior corpus callosum

The lesions correspond to those shown in Figure 14.19, except that the white matter that serves the left primary visual cortex is damaged, not the cortex itself. Source: (Based on Mao-Draayer, Y., & Panitch, H. (2004). Alexia without agraphia in multiple sclerosis: case report with magnetic resonance imaging localization. *Multiple Sclerosis Journal*, 10(6), 705–707.)

Toward an Understanding of Reading (1 of 5)

Role of the Brain

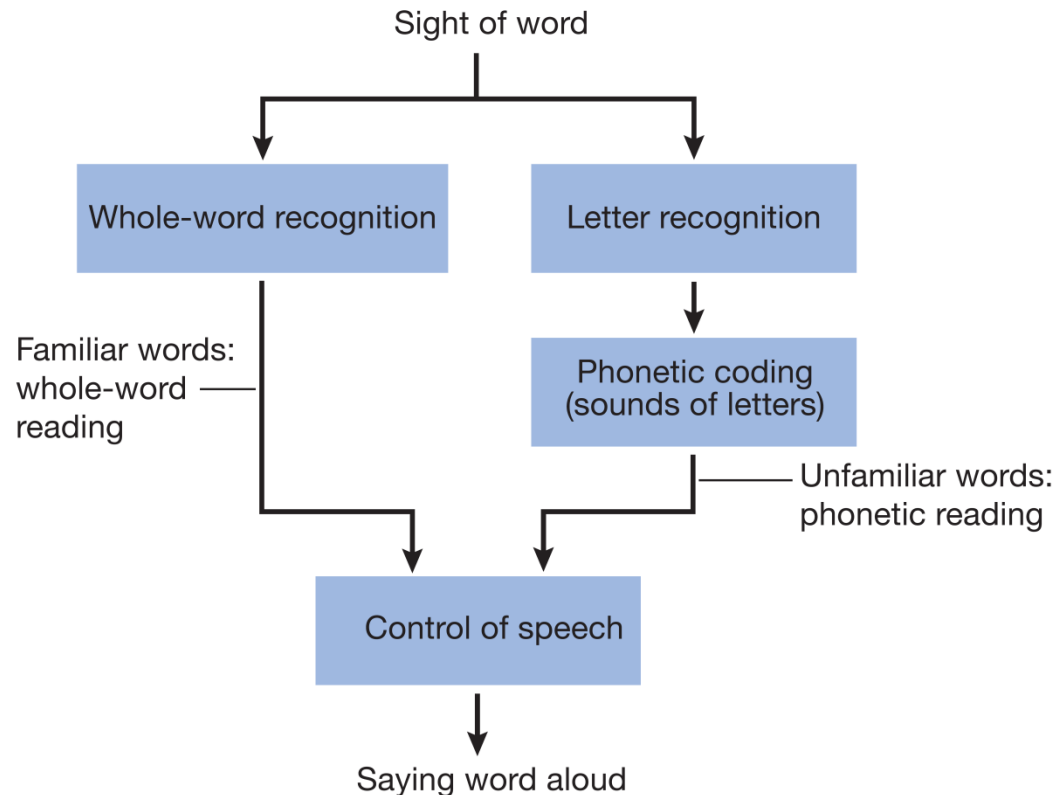
- Reading involves two processes
 - **Whole-word reading**
 - **Phonetic reading**

Toward an Understanding of Reading (1 of 5)

- Study of dyslexias provided information about brain mechanisms involved in reading
 - Acquired dyslexia
 - Developmental dyslexia

Figure 14.21

Model of the Reading Process



In this simplified model, whole-word reading is used for most familiar words; phonetic reading is used for unfamiliar words and for nonwords such as glab, trisk, or chint.

Toward an Understanding of Reading (2 of 5)

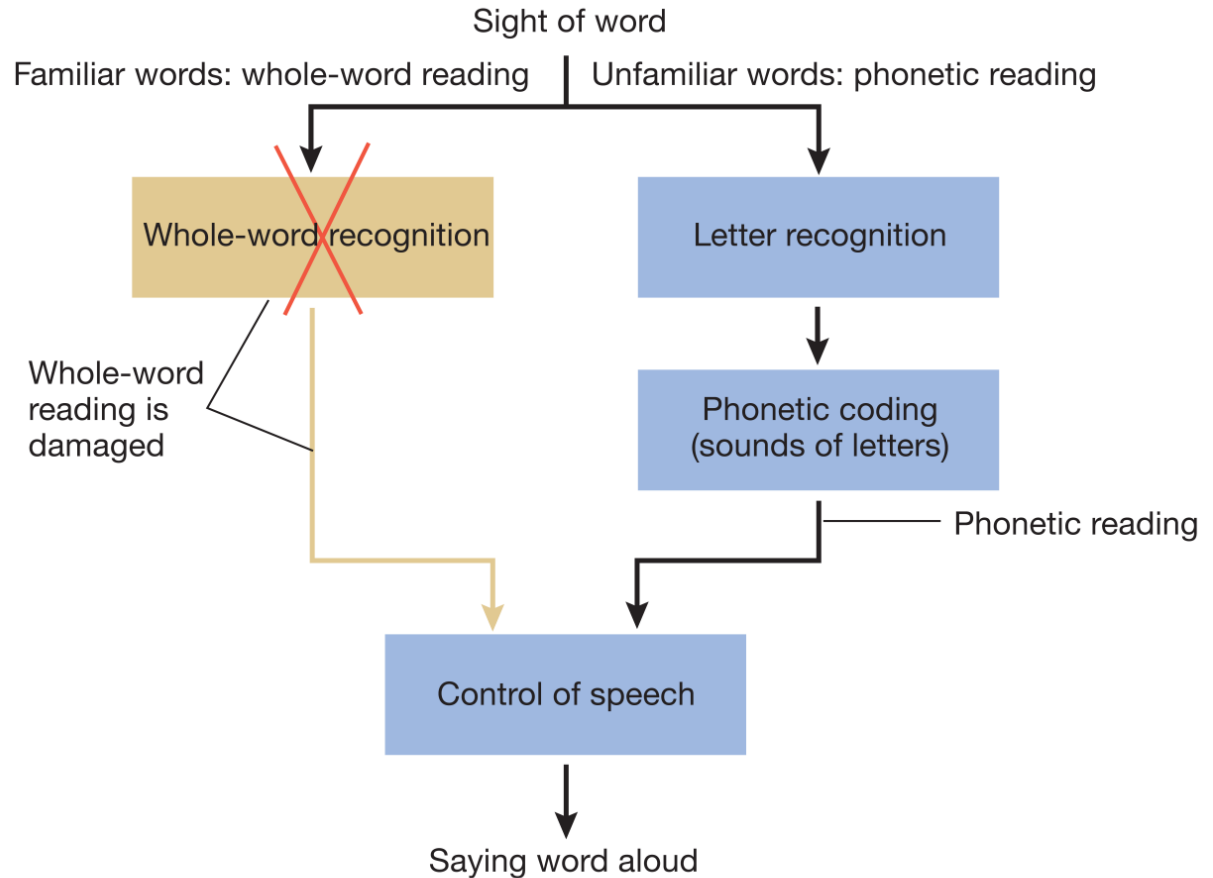
What are the differences between these type of Acquired Dyslexias?

- **Surface dyslexia –**
- **Direct dyslexia-**
- **Phonological dyslexia –**

Acquired Dyslexias:

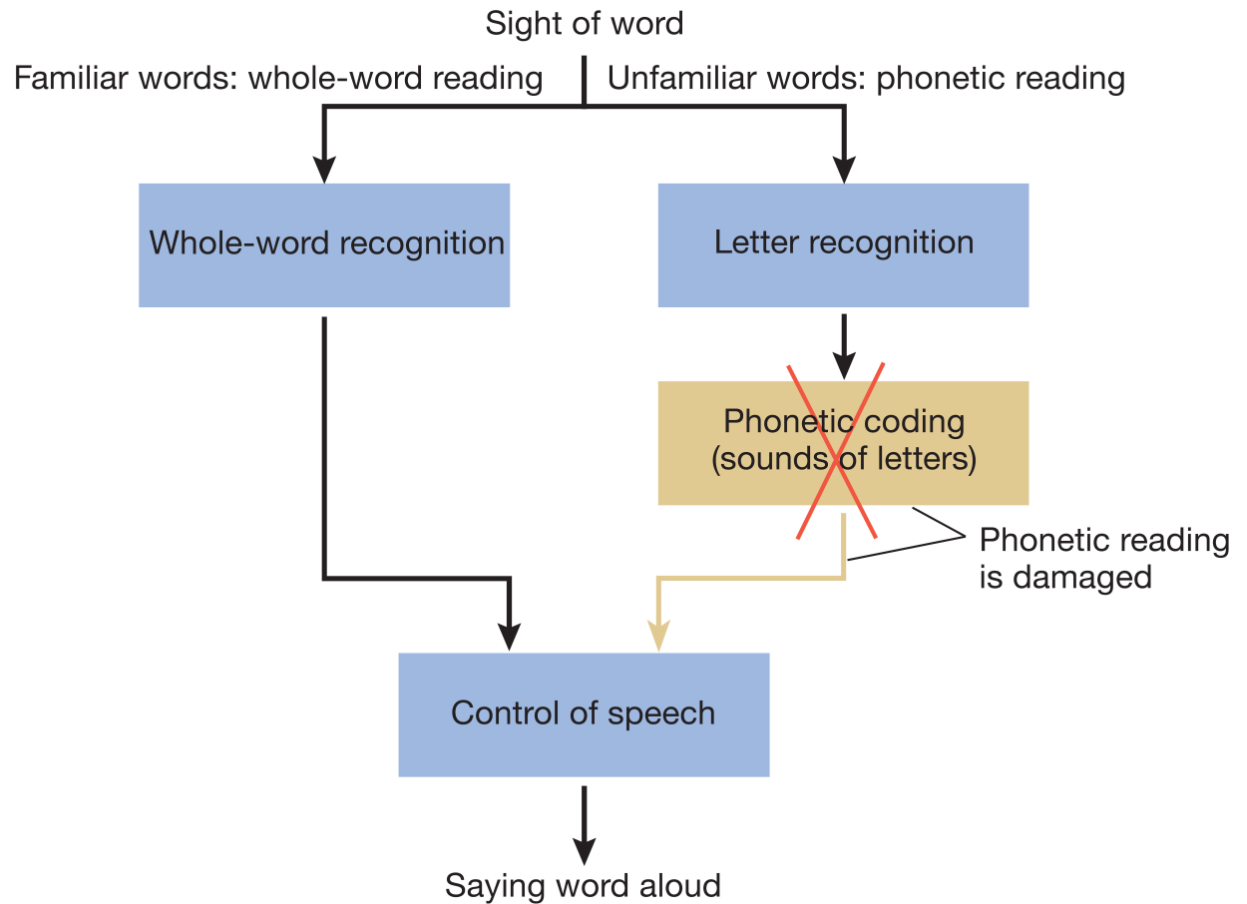
- **Surface dyslexia** – deficit in whole-word reading
- **Direct dyslexia** – can read words aloud but have deficit in understanding written words
 - Like transcortical sensory aphasia
- **Phonological dyslexia** – deficit in sounding words out
 - Studies of kana and kanji symbols

Figure 14.22 Surface Dyslexia



In this hypothetical example, whole-word reading is damaged; only phonetic reading remains.

Figure 14.23 Phonological Dyslexia



In this hypothetical example, phonetic reading is damaged; only whole-word reading remains.

Kanji-whole words and Kana-phonetic representations of syllables

Kanji: Represent whole words.

時 木 林

Time

This is the Kanji symbol for a tree. The symbol strongly suggests the shape of a tree.

This is the Kanji symbol for a forest. It suggests trees close together.

Kana: Represent sounds (not whole words).

Hiragana					Katakana				
あ	い	う	え	お	ア	イ	ウ	エ	オ
a	i	u	e	o	a	i	u	e	o
か	き	く	け	こ	カ	キ	ク	ケ	コ
ka	ki	ku	ke	ko	ka	ki	ku	ke	ko
さ	し	す	せ	そ	サ	シ	ス	セ	ソ
sa	shi	su	se	so	sa	shi	su	se	so
た	ち	つ	て	と	タ	チ	ツ	テ	ト
ta	chi	tsu	te	to	ta	chi	tsu	te	to
な	に	ぬ	ね	の	ナ	ニ	ヌ	ネ	ノ
na	ni	nu	ne	no	na	ni	nu	ne	no
は	ひ	ふ	へ	ほ	ハ	ヒ	フ	ヘ	ホ
ha	hi	hu	he	ho	ha	hi	hu	he	ho
ま	み	む	め	も	マ	ミ	ム	メ	モ
ma	mi	mu	me	mo	ma	mi	mu	me	mo
や		ゆ		よ	ヤ		ユ		ヨ
ya		yu		yo	ya		yu		yo
ら	り	る	れ	ろ	ラ	リ	ル	レ	ロ
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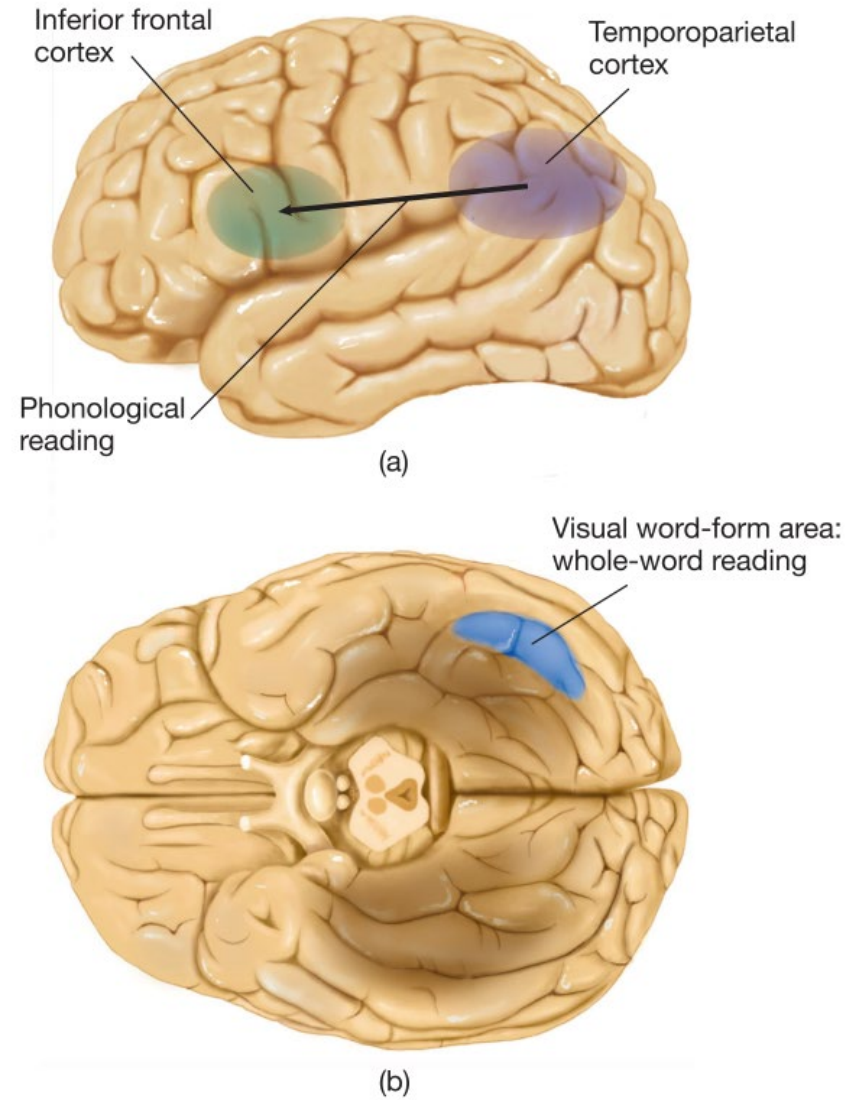
Visual Word-Form Area (VWFA)

- Plays a central role in both phonological and whole-word reading
- Parts of the visual association cortex must be involved in perceiving written words
- Damage to VWFA produces surface dyslexia and impaired whole-word reading

Figure 14.25

Phonological and Whole-Word Reading

(a) Phonological reading. (b) Whole-word reading.
VWFA = visual word-form area of the fusiform cortex.



a) Phonological reading. (b) Whole-word reading. VWFA = visual word-form area of the fusiform cortex.

Figure 14.26

Subtle Differences in Written Words

English	Arabic	Hindi	Mandarin
cars ears	رمان زمان pomegranate time/era	आज आजा today come	夫 天 man sky

Unless you can read Arabic, Hindi, or Mandarin, you will probably have to examine these words carefully to find the small differences. However, as a reader of English, you will immediately recognize the words car and ear. Source: (Devlin, J. T., Jamison, H. L., Gonnerman, L. M., & Matthews, P. M. (2006). The role of the posterior fusiform gyrus in reading. *Journal of cognitive neuroscience*, 18(6), 911–922.)

Plasticity in VWFA

- Written language is a recent invention
- Brain plasticity allowed development of language and writing systems across many cultures
 - Fluency and literacy can develop quickly
 - Importance of constancy in visual system, such as ways lines meet at vertices (example: letters L, T, X)

Human Communication

- Humans have been able to communicate for thousands of years.
 - However, each new generation started at the same baseline, without written records to refer to.
- The earliest book that we know of is a Chaldean account of the Flood.
 - On a clay tablet about 4000 B.C., antedating Genesis by some 2,000 years
- The first journal article was published 350 years ago.
- IMRAD was developed within the past century

Human Communication

Writing emerged around 3200 BCE in Mesopotamia (cuneiform) and Egypt (hieroglyphs)

- Initially used for record-keeping, trade, and administration
- Enabled preservation and transmission of knowledge across generations

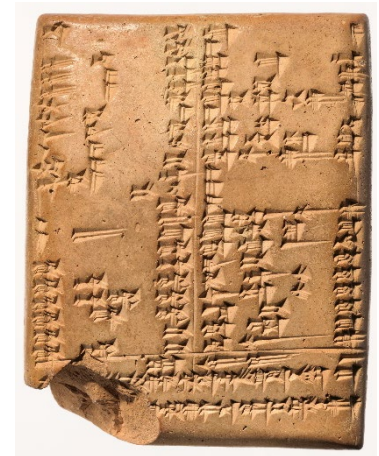
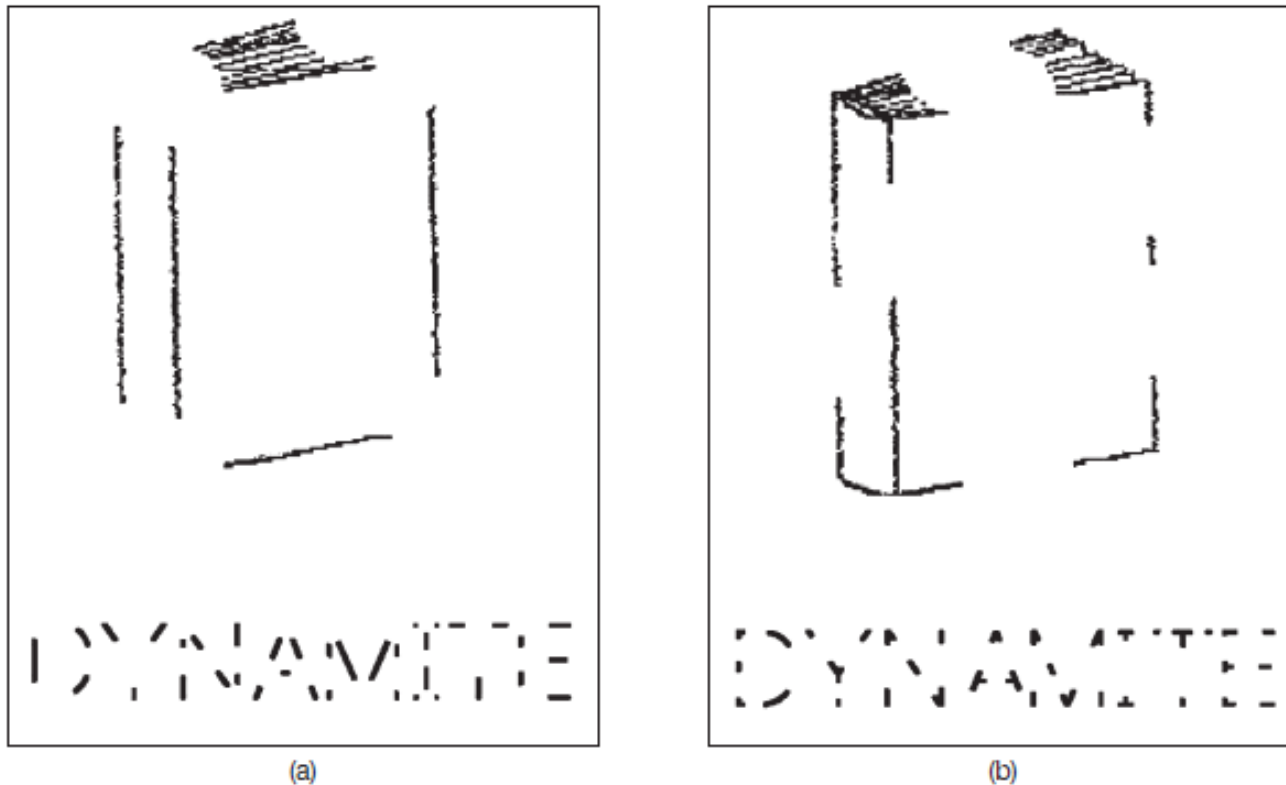


Figure 14.27

Object Recognition



a) Object and word recognition after vertices have been eliminated. (b) Object and word recognition after midsegments of lines have been eliminated. Source: (Based from Szwed, M., Cohen, L., Qiao, E., & Dehaene, S. (2009). The role of invariant line junctions in object and visual word recognition. *Vision research*, 49(7), 718–725.)

dyslexia.yale.edu/dyslexia/dyslexia-faq/



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Dyslexia FAQ

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What is dyslexia?
+

How common is dyslexia?
-

Dyslexia affects 20 percent of the population and represents 80–90 percent of all those with learning disabilities. It is the most common of all neuro-cognitive disorders.

Can smart people be dyslexic?
+

Can people with dyslexia learn to read?
+

Share

Dyslexia

What is Dyslexia?

Dyslexia FAQ



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Developmental Dyslexias

- Genetic component
 - Monozygotic twins-84-100%
 - Dizygotic-20-35%
 - Chromosome 1,3,6, and 15
- Involves more than deficits in reading
 - Phonological impairments
 - Difficulty in writing

Toward an Understanding of Reading (5 of 5)

- Reading skill is positively associated with activation of VWFA
- Decreased activation in left temporoparietal cortex and in VWFA for dyslexics

Toward an Understanding of Reading (5 of 5)

- Hyperactivation of left inferior frontal cortex and Broca's area
 - Maybe effort to decode the phonology of incomplete information from poorly functioning regions involved with reading

International differences in dyslexia

- Low rate of dyslexia for those in Italy, Spanish, Czech, and Japanese
- English has 40 sounds and words can be spelled in 1,120 different ways
- So many irregular words: cough, rough, bough, and through

International differences in dyslexia

- Italy has 25 different sounds that can be spelled in only 33 combinations of letters
- One Australian boy who lived in Japan and learned to read Japanese, experienced dyslexia in English

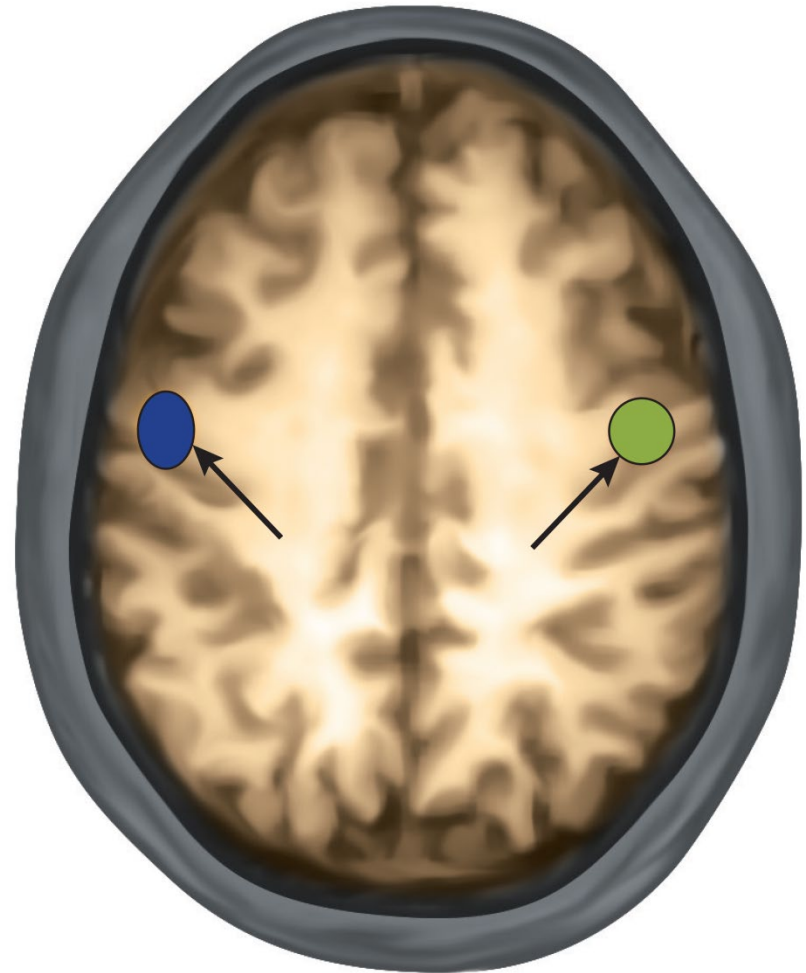
Dysgraphia

- Dysgraphia is a writing disturbance, common with dyslexia
- Many brain regions involved
 - Motor aspects involve the dorsal parietal cortex and premotor cortex
- Brain damage can disrupt ability to form letters or impair ability to spell

Figure 14.29

Writing and the Ventral Premotor Cortex

Subjects viewing letters activated the ventral premotor cortex in the hemisphere used for writing: the left hemisphere in right-handed subjects (blue) and the right hemisphere in left-handed subjects (green). Source: (Based on Longcamp, M., Anton, J. L., Roth, M., & Velay, J. L. (2005). Premotor activations in response to visually presented single letters depend on the hand used to write: a study on left-handers. *Neuropsychologia*, 43(12), 1801–1809.)



Toward an Understanding of Writing (2 of 3)

Skills Used in Writing

- Audition
- Vision
- Memorization
- Motor memory

Neural Basis of Writing

Brain damage can impair writing

- **Phonological dysgraphia**
 - Unable to sound out words and write them phonetically
 - Cannot write unfamiliar words or pronounceable words
 - Can visually imagine familiar words and then write them

Neural Basis of Writing

Orthographic dysgraphia

- Can only sound words out
- Can spell out regular words like tree or care
- Can write pronounceable nonsense words
- Have difficulty spelling out irregular words like half or busy

Table 14.3 Brain Regions Involved in Disorders of Reading and Writing (1 of 2)

Reading Disorder	Whole-Word Reading	Phonetic Reading	Region of Brain Damage
Pure alexia (can write)	Poor	Poor	Left primary visual cortex and posterior corpus callosum
Surface dyslexia	Poor	Good	Visual word-form area
Phonological dyslexia	Good	Poor	VWFA. Others unknown; likely temporoparietal cortex and inferior frontal cortex (including Broca's area).

Table 14.3 Brain Regions Involved in Disorders of Reading and Writing (2 of 2)

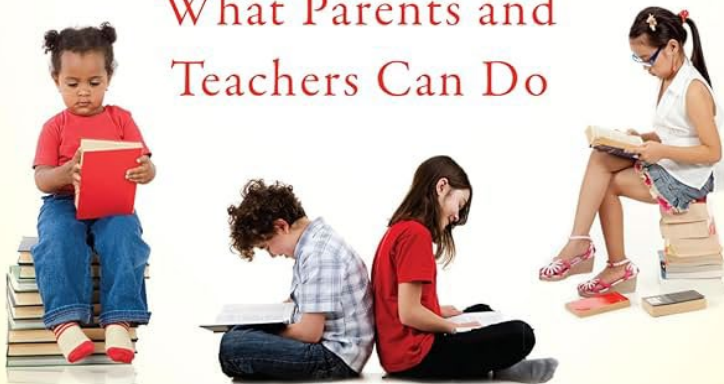
Writing Disorder	Whole-Word Writing	Phonetic Writing	Region of Brain Damage
Phonological dysgraphia	Good	Poor	Broca's area, ventral precentral gyrus, insula
Orthographic dysgraphia	Poor	Good	Posterior inferior temporal cortex

DANIEL T. WILLINGHAM, PhD

Author of *Why Don't Students Like School?*

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What Parents and
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IS HARD
AND HOW YOU CAN
MAKE IT EASY



Daniel T.
Willingham, PhD

Bestselling author of *Why Don't Students Like School?*

Learning Objectives (1 of 4)

- 14.1** Compare the language-related functions of the left and right hemispheres, and discuss the prevalence of these functions in the opposite hemisphere.
- 14.2** List factors involved in language production, and identify corresponding brain regions.
- 14.3** Describe the factors involved in language comprehension, and identify corresponding brain regions.
- 14.4** Describe evidence for common and language-specific brain regions for bilingual language processing.

Learning Objectives (2 of 4)

- 14.5** Identify brain structures and functions involved in prosody.
- 14.6** Identify the brain regions involved in recognizing people's voices.
- 14.7** Describe the symptoms and neural mechanisms of Broca's aphasia, including agrammatism, anomia, and articulation.
- 14.8** Describe the symptoms and neural bases of Wernicke's aphasia, including deficits in spoken word recognition, comprehension, and converting thoughts into words.

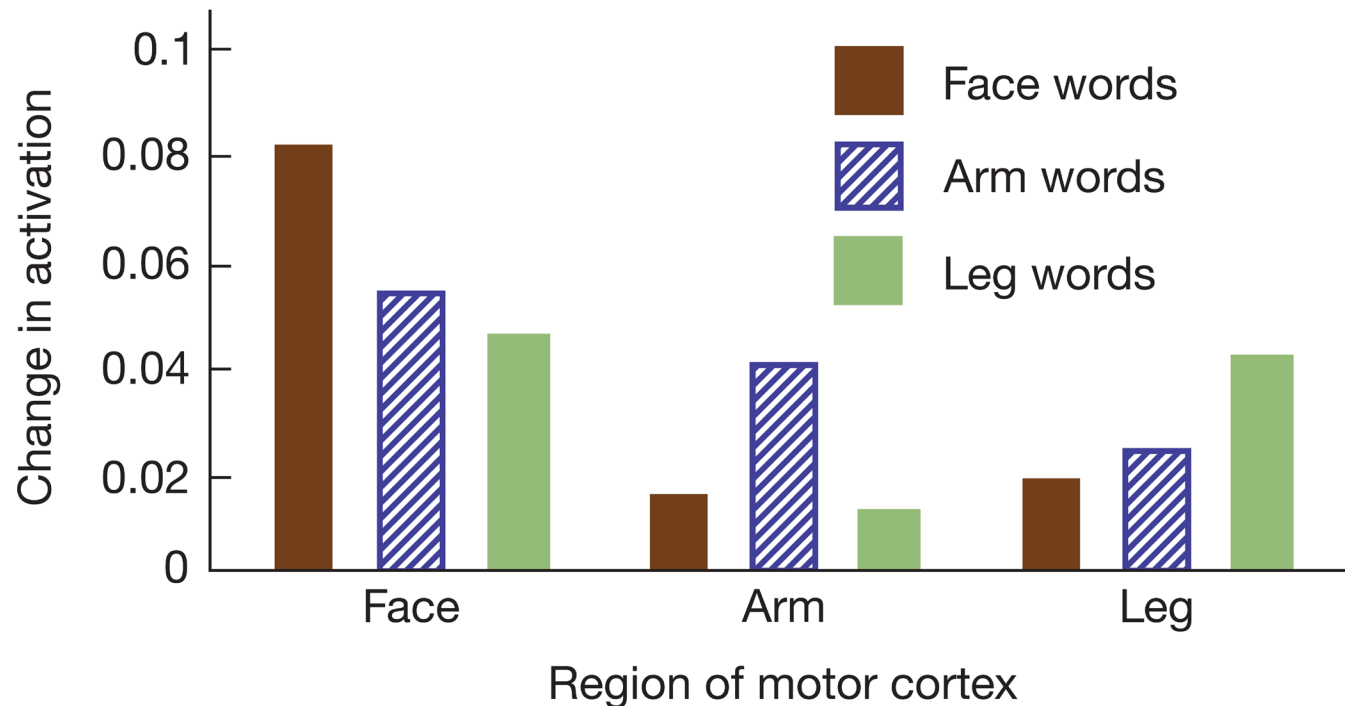
Learning Objectives (3 of 4)

- 14.9** Identify the symptoms of conduction aphasia, and describe their contribution to understanding about connections between Broca's and Wernicke's areas.
- 14.10** Explain the brain processes involved in production and comprehension of American Sign Language.
- 14.11** Describe the biological basis for stuttering and identify treatment strategies.
- 14.12** Compare aphasias and disorders of reading and writing.

Learning Objectives (4 of 4)

- 14.13** Identify the symptoms of pure alexia, and describe the brain structures involved.
- 14.14** Describe how research on acquired and developmental dyslexia helps us to understand the role of the brain in reading.
- 14.15** Explain how research on audition, vision, memorization, and motor control helps us to understand the role of the brain in writing.

Figure 14.11 Verbs and Movements



The figure shows the relative activation of regions of the motor cortex that control movements of the face, arm, and leg, when people read verbs that described movements of these regions, such as bite, slap, and kick. Source: (Based on Hauk, O., Johnsrude, I., & Pulvermüller, F. (2004). Somatotopic representation of action words in human motor and premotor cortex. *Neuron*, 41(2), 301–307.)

Factors in Language Production

- Individual must have something to communicate: current event, past event (memories) or imagined event
- Occipital, temporal, and parietal lobes of brain are involved in perceptions of current and past events
- Little is known about brain mechanisms involved in imagination, but likely use same areas

Language Comprehension

Factors in Comprehension

- Recognizing words may be auditory or visual
- Comprehension involves recognition of a word using Wernicke's area in the left temporal lobe
 - Must also understand the meanings of words and memories associated with them